On the Other Side of the Creek

Historical Exclusion, Local Interdependence, and Human Capital Gaps

L. Guillermo Woo-Mora
(PSE and EHESS)

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Social exclusion can turn social life inward

- **Exclusion**: denying a group the standing and access that others in the same society enjoy (Sen, 2000)
  - **Material**: fewer schools, services, and infrastructure; less access to land and credit
  - **Social**: lower status, stigma, and being kept out of networks

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- So they turn to each other for help, care, information, and insurance (Bowles and Gintis, 2004)

Group benefits can compete with education

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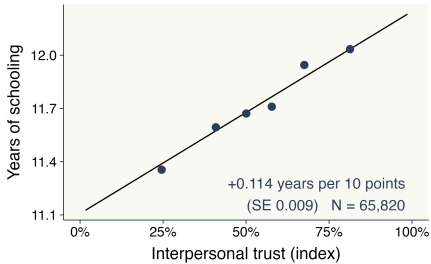

Broader trust, more schooling

World Values Survey, 42 countries; within-country comparisons

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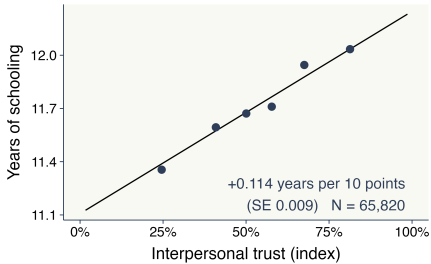

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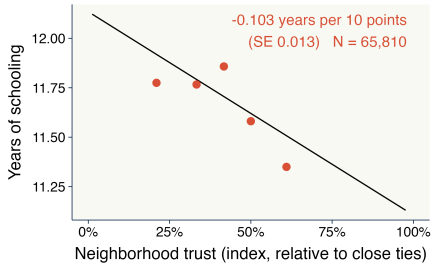

Trust centered on neighbors, less schooling

World Values Survey, 42 countries; within-country comparisons

But which way does it run?

This paper

In a place shaped by historical exclusion,
does the neighborhood affect the demand for education?

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In a place shaped by historical exclusion,
does the neighborhood affect the demand for education?

- **Setting**: a historical natural experiment in neighborhood formation in Guadalajara, Mexico (1542)
  - conquerors and their allies settled **west** of a creek; the conquered, **east**
  - **East = historically excluded**: **socially**, lower standing; **materially**, fewer public goods
  - same jurisdiction and geography; the material gap has since closed

This paper

In a place shaped by historical exclusion,
does the neighborhood affect the demand for education?

- **Setting**: a historical natural experiment in neighborhood formation in Guadalajara, Mexico (1542)
- **Persistence**: measure the educational gap at the former creek, now an avenue, 1930–2020
  - spatial RDD with census microdata
  - test standard explanations on the supply side: school supply and quality, neighborhood amenities
  - and on the demand side: parental schooling, household resources, returns to schooling

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- **Persistence**: measure the educational gap at the former creek, now an avenue, 1930–2020
- **Pre-registered experiment**: parents' demand for education, with and without the neighborhood in mind
  - demand: incentivized choice, math tutoring vs. a gift card
  - prime: at random, parents map their neighborhood just before or just after choosing

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- **Interpretation**: surveys, historical records, and a residential shock help interpret both results
- **Framework**: the return to schooling vs. the neighbor exchange it puts at risk
  - static: explain gaps and experiment; dynamic: why the gap persists

Preview of findings

Preview of findings

- **Persistence: the colonial boundary still predicts human capital**
  - ↓ 0.8 fewer years of schooling on the historically excluded east side
  - ↓ school attendance among the school-age population
    - 10.6 pp lower in 1930
    - 3–6 pp lower since 2000, while school provision converged
    - largest in adolescence, around the move to high school
    - at most a third explained by parental schooling

Preview of findings

- **Persistence: the colonial boundary still predicts human capital**
- **Experiment: bringing the neighborhood to mind lowers the demand for education**
  - east side: ↓ 12 pp fewer tutoring choices (a fifth); ↓ MXN 75 lower WTP (two fifths)
  - west side: no detectable effect

Preview of findings

- **Persistence: the colonial boundary still predicts human capital**

- **Experiment: bringing the neighborhood to mind lowers the demand for education**

- **Interpretation: what differs between west and east**

  East side: more exchange with neighbors, not a lower valuation of education

  - ↑ trust in and help from neighbors; ↓ trust in institutions
  - same aspirations and perceived returns, but a higher bar for university
  - historical records: locally organized social life since at least 1821
  - residential shock: positive selection, then a return to the neighborhood's level

Preview of findings

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Where exclusion turned social life inward, the neighborhood competes with education

Contributions to the literature

1. **Networks, social interactions, and educational investment** (Coleman, 1988; Munshi, 2014, 2020; Breza et al., 2019)
   - demand for education: parents' human capital, credit, returns (Becker and Tomes, 1986; Cunha and Heckman, 2007; Galor and Zeira, 1993; Galor, 2011)
   - networks shape schooling, migration, and work; groups taken as given: caste, kin, ethnicity (Munshi and Rosenzweig, 2006, 2016; Carranza et al., 2025)
   - networks benefits fade with outside options (Kranton, 1996; Banerjee and Newman, 1998; Dixit, 2003; Banerjee et al., 2024)

$\Rightarrow$ **Groups need not be inherited: the neighborhood can act as one, and it shapes the demand for education**

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2. **Historical development and persistence** (Nunn, 2014, 2020; Voth, 2021)
   - history persists through institutions (Acemoglu et al., 2001; Dell, 2010), geography (Davis and Weinstein, 2002; Bleakley and Lin, 2012), human capital (Valencia-Caicedo, 2019), and culture (Nunn, 2012; Michalopoulos and Xue, 2021; Lowes, 2023)
   - history shapes the layout of cities (Ambrus et al., 2020; Heblich et al., 2021; Hanlon and Heblich, 2022; Lin and Rauch, 2022; Baldomero-Quintana et al., 2025)

   ⇒ **Within one city, exclusion's educational gap outlives measured material convergence**
   - exclusion leaves lasting distrust (Nunn and Wantchekon, 2011; Grosfeld and Zhuravskaya, 2015; Alsan and Wanamaker, 2018; Lowes and Montero, 2021; Ramos-Toro, 2023)

   ⇒ **The flip side of in-group trust: neighbor ties compete with education**

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3. **Neighborhood effects** (Durlauf, 2004; Topa and Zenou, 2015; Chyn and Katz, 2021; Katz, 2026)
   - where children grow up shapes their outcomes (Kling et al., 2007; Ludwig et al., 2013; Chetty et al., 2016; Bergman et al., 2024)
   - partly through the social environment (Sampson et al., 2002; Chetty et al., 2026)
   - moving severs local ties (Barnhardt et al., 2017; Franklin, 2020; Belchior et al., 2023; Carrillo et al., 2023; Franklin, 2025; Bryan et al., 2026; Rojas-Ampuero and Carrera, 2026)

   ⇒ **A social channel of neighborhood effects, revealed without moving anyone**

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Introduction

Context and Historical Natural Experiment

Persistent Human Capital Gaps – Spatial Rdd

The Neighborhood and the Demand for Education - Experiment

Conclusion

Context and Historical Natural Experiment

Guadalajara: Mexico's second-largest metropolitan area, unequal within

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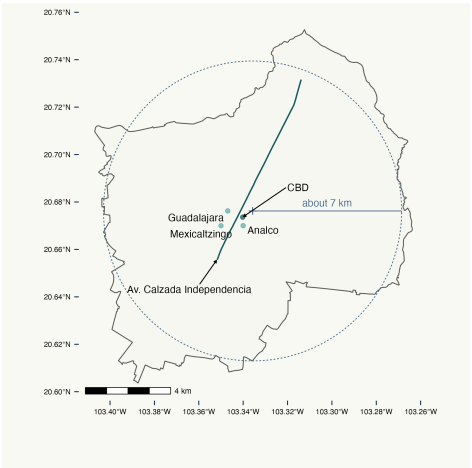
- **Guadalajara** anchors Mexico's **second-largest metropolitan area**
- Relatively rich and schooled: income **36% above** the national average

National income distribution | School attendance, 1970–2020

Guadalajara: Mexico's second-largest metropolitan area, unequal within



Municipality of Guadalajara: **1.39 million** residents

- **Guadalajara** anchors Mexico's **second-largest metropolitan area**
- Relatively rich and schooled: income **36% above** the national average

  National income distribution | School attendance, 1970–2020
- Yet large educational differences **within cities**
  - across Mexico, **69%** of the variance in attendance is unexplained variation between households in the same municipality; only **4%** is between municipalities

Variance decomposition | School supply | Test scores

Guadalajara, 1542: colonial standing changes across a creek

- A city founded from scratch: **no pre-existing settlements**
- **West**: Spaniards and, in **Mexicaltzingo**, their Indigenous allies
  "not conquered, but among the conquerors"
- **East**: **Analco**, for the conquered

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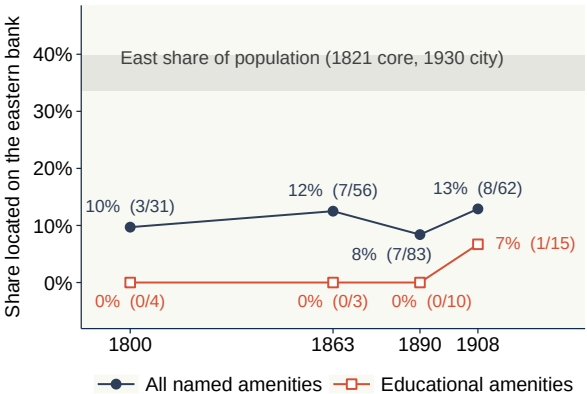
  "not conquered, but among the conquerors"
- **East**: **Analco**, for the conquered
- **Same jurisdiction and geography, different colonial or social standing**

No precolonial agglomerations | Same locational fundamentals | Same jurisdiction | 1821 census

Exclusion was material



Named institutions by bank, city maps 1800–1908

- East: up to **two-fifths** of the population, **under 15%** of amenities

Maps and categories | 1882 land values

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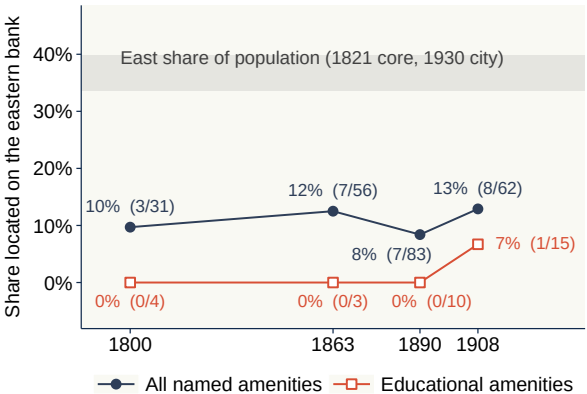


Named institutions by bank, city maps 1800–1908

- East: up to **two-fifths** of the population, **under 15%** of amenities
- **No school** east of the creek until 1908

Maps and categories | 1882 land values

From a creek to an avenue: the line outlived its physical form

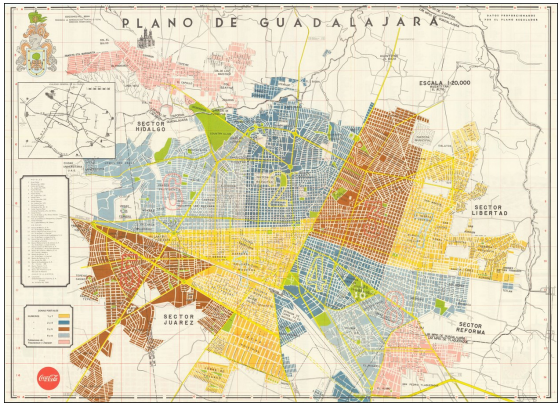
Guadalajara, 1908

From a creek to an avenue: the line outlived its physical form

Piping the creek, 1911

- Piped underground in 1903–1911 and replaced by an avenue The avenue today

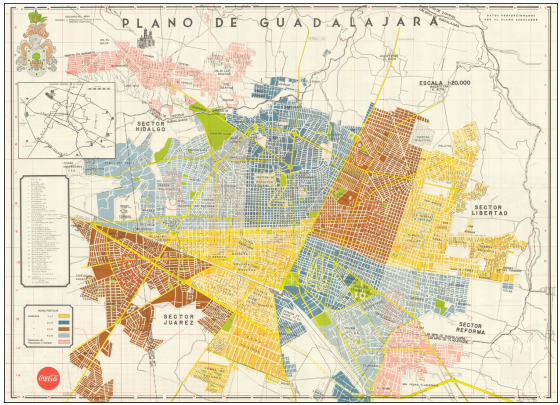
From a creek to an avenue: the line outlived its physical form



Guadalajara, 1972

- Piped underground in 1903–1911 and replaced by an avenue The avenue today
- **Not a barrier**: **36 m** wide, with a street crossing every **151 m** Crossings
- **Not an ethnic line**: 0.3% speak an Indigenous language; skin tone dispersed Skin tone

From a creek to an avenue: the line outlived its physical form



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- **Yet the avenue remains a social marker**: *de la Calzada para allá*
  "from the Calzada over there": a name for the east side, carrying stigma since the early 1900s

(Trujillo Bretón, 2018; Reguillo, 1998; Jaramillo-Molina and Saucedo, 2016; Jaramillo-Molina, 2022)

Persistent Human Capital Gaps

Spatial RDD: comparing residents on either side of the former creek

- **Data:** 2020 Census, full count, geolocated to the block

- **Specification:** spatial RDD or boundary design

$$y_i = \alpha + \tau\, \mathbb{1}\{East_i\} + \delta s_i + \zeta\, \mathbb{1}\{East_i\} s_i + \gamma' X_i + \varepsilon_i$$

  - $\mathbb{1}\{East_i\}$: lives east of the former creek, the colonial Analco side
  - s_i: signed distance to the boundary, with side-specific slopes
  - X_i: age, sex, coordinates, boundary-segment FE

- **Best practice** (Calonico et al., 2014; Cattaneo et al., 2026)
  side-specific slopes ✓ MSE-optimal h ✓ bias-corrected CIs ✓ $\tau(b)$ along the line ✓

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  - same locational fundamentals, jurisdiction, and labor market; no school catchments
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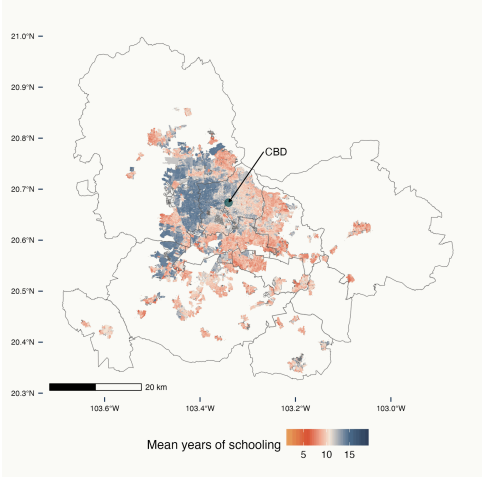
  - τ is the **total long-run effect**: status, provision, transmission, and sorting

Human capital drops at the historical boundary



Urban census blocks, 2020. White: unpopulated, or values suppressed by INEGI for confidentiality

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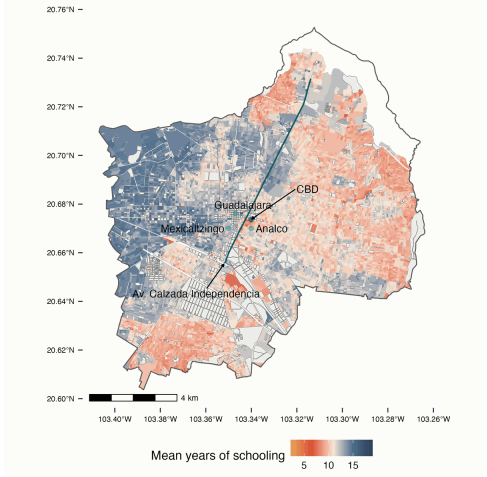

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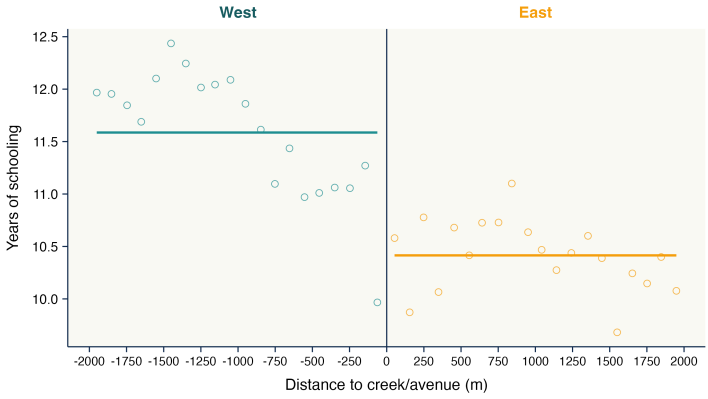

Side means within 2 km: about one year apart

Years of schooling, ages 6+; 100 m bins

All specifications | Balance | Migrants | Donut | Placebo boundaries

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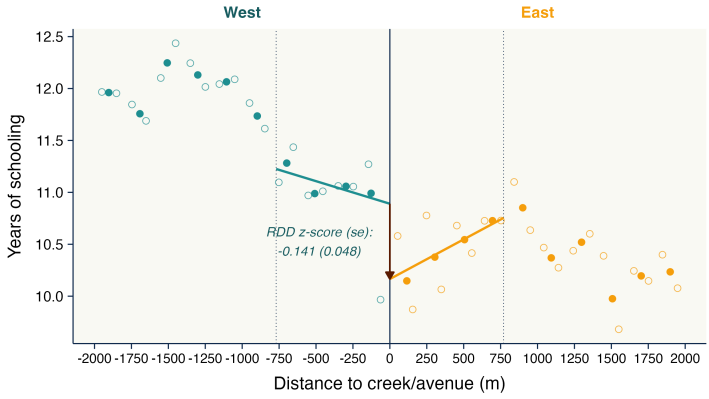

0.8 fewer years of schooling (0.14–0.17 SD)

0.17 SD: no covariates; 0.14 SD: with covariates. Local linear fits within the bias-correction window (±770 m)

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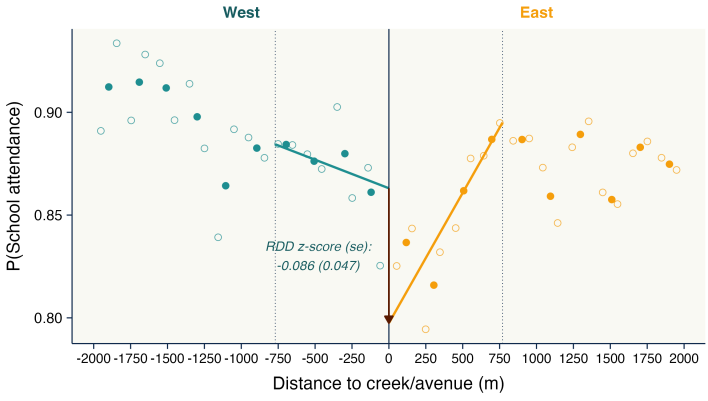

School attendance 3.2 pp lower on the east side

3.7 pp comparing children whose parents have similar schooling

Ages 6–22. Raw: 6.0 pp → individual controls: 3.2 pp → + household controls: 3.7 pp

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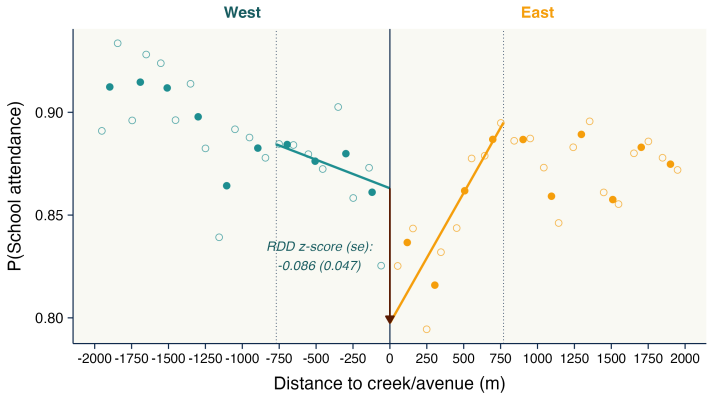
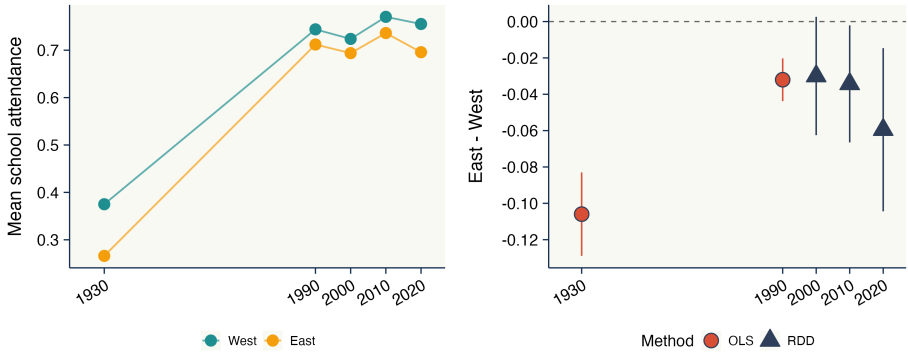


Largest in the historical core

facing both Spaniards and Indigenous allies: not ethnicity, not the avenue

All specifications | Balance | Migrants | Donut | Placebo boundaries | Along the boundary

The gap narrowed over the century, but has not closed



10.6 pp lower in 1930, before mass schooling; 3–6 pp since 2000, with no detectable change

1930: *cuarteles*, age and sex FE; 1990: tracts; 2000–2020: block RDD, no controls

1930 census | Years of schooling | Literacy

Parental schooling explains at most a third of the gap

2015 Intercensal Survey:

east–west gap with side-specific linear trends in distance

	Years of schooling
	(1)
$\mathbb{1}\{East_i\}$	-1.31***
	(0.118)
Years of schooling (YoS)	
$\mathbb{1}\{East_i\} \times$ YoS	
Parental YoS	
Household income (log)	
Sample (age in years)	6+
Individual controls	✓
Geographic controls	✓
Household size	
Observations	74,779
Gap relative to mean (%)	-14.3
Dep. var. mean	9.18
Dep. var. SD	4.71
R^2	0.443

Gap relative to mean: 100× coefficient / dep. var. mean; log outcomes: coefficient already relative
Municipality-wide with linear distance trends; the boundary RDD gives 0.8 years

Stock: 1.3 fewer years of schooling on the east side

Household resources | Credit and housing finance

Parental schooling explains at most a third of the gap

2015 Intercensal Survey:

east–west gap with side-specific linear trends in distance

	Years of schooling	$\mathbb{1}\{Attends\ school_i\}$	
	(1)	(2)	(3)
$\mathbb{1}\{East_i\}$	-1.31***	-0.052***	-0.081***
	(0.118)	(0.015)	(0.024)
Years of schooling (YoS)			
$\mathbb{1}\{East_i\} \times$ YoS			
Parental YoS			
Household income (log)			
Sample (age in years)	6+	6–18	12–18
Individual controls	✓	✓	✓
Geographic controls	✓	✓	✓
Household size			
Observations	74,779	17,463	9,922
Gap relative to mean (%)	-14.3	-5.9	-10.0
Dep. var. mean	9.18	0.884	0.807
Dep. var. SD	4.71	0.320	0.395
R^2	0.443	0.181	0.127

Gap relative to mean: 100× coefficient / dep. var. mean; log outcomes: coefficient already relative
Municipality-wide with linear distance trends; the boundary RDD gives 0.8 years

Flow: largest attendance gap at ages 12–18 (-8 pp)

Household resources | Credit and housing finance

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east–west gap with side-specific linear trends in distance

	Years of schooling	$\mathbb{1}\{Attends\ school_i\}$				
	(1)	(2)	(3)	(4)	(5)	(6)
$\mathbb{1}\{East_i\}$	-1.31***	-0.052***	-0.081***	-0.054**	-0.051**	-0.052**
	(0.118)	(0.015)	(0.024)	(0.024)	(0.023)	(0.025)
Years of schooling (YoS)						
$\mathbb{1}\{East_i\} \times$ YoS						
Parental YoS				0.021***	0.018***	0.019***
				(0.001)	(0.001)	(0.001)
Household income (log)						0.0002
						(0.008)
Sample (age in years)	6+	6–18	12–18	12–18	12–18	12–18
Individual controls	✓	✓	✓	✓	✓	✓
Geographic controls	✓	✓	✓	✓	✓	✓
Household size					✓	✓
Observations	74,779	17,463	9,922	9,916	9,916	9,041
Gap relative to mean (%)	-14.3	-5.9	-10.0	-6.7	-6.3	-6.5
Dep. var. mean	9.18	0.884	0.807	0.807	0.807	0.803
Dep. var. SD	4.71	0.320	0.395	0.395	0.395	0.398
R^2	0.443	0.181	0.127	0.172	0.183	0.193

Gap relative to mean: 100× coefficient / dep. var. mean; log outcomes: coefficient already relative
Municipality-wide with linear distance trends; the boundary RDD gives 0.8 years

Parental schooling: $-8 \rightarrow -5$ pp; household size and income: no change

Household resources | Credit and housing finance

Parental schooling explains at most a third of the gap

2015 Intercensal Survey:

east–west gap with side-specific linear trends in distance

	Years of schooling	$\mathbb{1}\{Attends\ school_i\}$					
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
$\mathbb{1}\{East_i\}$	-1.31***	-0.052***	-0.081***	-0.054**	-0.051**	-0.052**	-0.015*
	(0.118)	(0.015)	(0.024)	(0.024)	(0.023)	(0.025)	(0.008)
Years of schooling (YoS)							
$\mathbb{1}\{East_i\} \times$ YoS							
Parental YoS				0.021***	0.018***	0.019***	0.003***
				(0.001)	(0.001)	(0.001)	(0.0006)
Household income (log)						0.0002	0.0000
						(0.008)	(0.002)
Sample (age in years)	6+	6–18	12–18	12–18	12–18	12–18	6–11
Individual controls	✓	✓	✓	✓	✓	✓	✓
Geographic controls	✓	✓	✓	✓	✓	✓	✓
Household size					✓	✓	✓
Observations	74,779	17,463	9,922	9,916	9,916	9,041	8,059
Gap relative to mean (%)	-14.3	-5.9	-10.0	-6.7	-6.3	-6.5	-1.5
Dep. var. mean	9.18	0.884	0.807	0.807	0.807	0.803	0.984
Dep. var. SD	4.71	0.320	0.395	0.395	0.395	0.398	0.127
R^2	0.443	0.181	0.127	0.172	0.183	0.193	0.020

Gap relative to mean: 100× coefficient / dep. var. mean; log outcomes: coefficient already relative
Municipality-wide with linear distance trends; the boundary RDD gives 0.8 years

Ages 6–11: -1.5 pp; the gap opens in adolescence

Household resources | Credit and housing finance

Parental schooling explains at most a third of the gap

2015 Intercensal Survey:

east–west gap with side-specific linear trends in distance

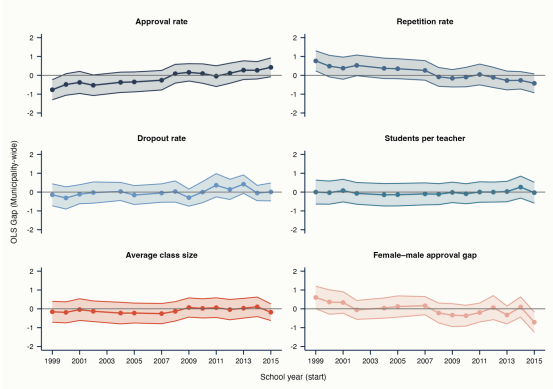
	Years of schooling	$\mathbb{1}\{Attends\ school_i\}$						Monthly income (log)		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
$\mathbb{1}\{East_i\}$	-1.31***	-0.052***	-0.081***	-0.054**	-0.051**	-0.052**	-0.015*	-0.084***	-0.010	0.075
	(0.118)	(0.015)	(0.024)	(0.024)	(0.023)	(0.025)	(0.008)	(0.027)	(0.024)	(0.068)
Years of schooling (YoS)									0.065***	0.068***
									(0.001)	(0.005)
$\mathbb{1}\{East_i\} \times$ YoS										-0.006
										(0.006)
Parental YoS				0.021***	0.018***	0.019***	0.003***			
				(0.001)	(0.001)	(0.001)	(0.0006)			
Household income (log)						0.0002	0.0000			
						(0.008)	(0.002)			
Sample (age in years)	6+	6–18	12–18	12–18	12–18	12–18	6–11	18+	18+	18+
Individual controls	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Geographic controls	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Household size					✓	✓	✓			
Observations	74,779	17,463	9,922	9,916	9,916	9,041	8,059	31,584	31,478	31,478
Gap relative to mean (%)	-14.3	-5.9	-10.0	-6.7	-6.3	-6.5	-1.5	–	–	–
Dep. var. mean	9.18	0.884	0.807	0.807	0.807	0.803	0.984	8.65	8.65	8.65
Dep. var. SD	4.71	0.320	0.395	0.395	0.395	0.398	0.127	0.644	0.645	0.645
R^2	0.443	0.181	0.127	0.172	0.183	0.193	0.020	0.123	0.268	0.269

Gap relative to mean: 100× coefficient / dep. var. mean; log outcomes: coefficient already relative
Municipality-wide with linear distance trends; the boundary RDD gives 0.8 years

Earnings 8% lower, explained by schooling; same returns on both sides

Household resources | Credit and housing finance

Schools and neighborhood conditions do not jump at the boundary



Secondary schools, 1999–2015: east minus west across the municipality, in SD

- **Schools**: same supply; same quality since the mid-2000s
  RDD gaps in school supply | School records (F911) | Test scores
- **Neighborhood**: crime, street amenities, urban form, businesses
  Crime | Amenities | Urban form | Businesses
- **Housing**: same schooling gap with land-value controls
  Validation | By land use | Block-level RDD: Schooling

Sorting remains possible, as one channel of persistence

Taking stock

- **Human capital drops at the historical boundary**
- **The gap narrowed over the century, but has not closed**
- **Parents' schooling and local conditions explain only part of it**

Room for another margin of educational demand

The Neighborhood and the Demand for Education

The hypothesis comes from the field and from Mexican urban ethnography

- **Fieldwork**: conversations with residents east of the Calzada
  - residents attribute the difficulty of getting ahead to their *barrio* (Jaramillo-Molina, 2022)

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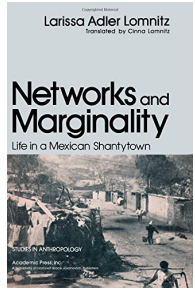
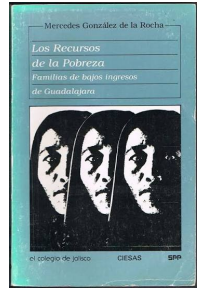


Mexico City, 1977



Guadalajara, 1994

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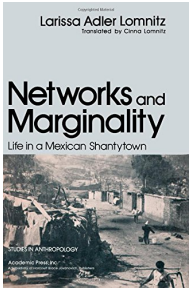

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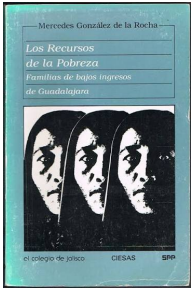

Guadalajara, 1994

Poor urban households get by through **reciprocal exchange with neighbors**

That exchange is costly to keep: **you must stay available and be able to give back**

Not a culture of poverty (Lewis, 1966): the cost is not lower values passed on regardless of opportunity, but the value of an exchange that depends on outside options.

Not a culture of poverty

City survey: more trust in neighbors east of the boundary

Life Quality Survey 2022, representative of Guadalajara: discontinuity at the boundary

	Trust in			
	Neighbors	Family	Coworkers or classmates	People on the street
	(1)	(2)	(3)	(4)
$\mathbb{1}\{East_i\}$	0.372**	0.072	0.134	0.049
	(0.149)	(0.110)	(0.158)	(0.129)
q-value	[0.035]	[0.390]	[0.329]	[0.481]
Observations	289	289	289	289
Dep. var. mean	0.655	0.912	0.585	0.217
Age–sex fixed effects	✓	✓	✓	✓
Survey weights	✓	✓	✓	✓

289 respondents, 23 tracts; side-specific distance gradients; robust SEs; sharpened q-values in brackets

37 pp more likely to trust neighbors; not family, coworkers, or strangers

Institutional trust | Memberships | Falsification

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Life Quality Survey 2022, representative of Guadalajara: discontinuity at the boundary

	Trust in				Institutional trust index	Member of neighborhood assoc.
	Neighbors	Family	Coworkers or classmates	People on the street		
	(1)	(2)	(3)	(4)	(5)	(6)
$\mathbb{1}\{East_i\}$	0.372**	0.072	0.134	0.049	-0.235***	0.066
	(0.149)	(0.110)	(0.158)	(0.129)	(0.064)	(0.056)
q-value	[0.035]	[0.390]	[0.329]	[0.481]	[0.004]	[0.268]
Observations	289	289	289	289	289	289
Dep. var. mean	0.655	0.912	0.585	0.217	0.229	0.041
Age–sex fixed effects	✓	✓	✓	✓	✓	✓
Survey weights	✓	✓	✓	✓	✓	✓

289 respondents, 23 tracts; side-specific distance gradients; robust SEs; sharpened q-values in brackets

Lower trust in institutions; no difference in memberships

Institutional trust | Memberships | Falsification

City survey: more trust in neighbors east of the boundary

Life Quality Survey 2022, representative of Guadalajara: discontinuity at the boundary

	Trust in				Institutional trust index	Member of neighborhood assoc.	Years of schooling
	Neighbors	Family	Coworkers or classmates	People on the street			
	(1)	(2)	(3)	(4)	(5)	(6)	(11)
$\mathbb{1}\{East_i\}$	0.372**	0.072	0.134	0.049	-0.235***	0.066	-0.798
	(0.149)	(0.110)	(0.158)	(0.129)	(0.064)	(0.056)	(1.23)
Trust in neighbors							0.796
							(0.691)
Trust in neighbors $\times$ $\mathbb{1}\{East_i\}$							-1.74*
							(0.945)
q-value	[0.035]	[0.390]	[0.329]	[0.481]	[0.004]	[0.268]	
Observations	289	289	289	289	289	289	289
Dep. var. mean	0.655	0.912	0.585	0.217	0.229	0.041	11.33
Age–sex fixed effects	✓	✓	✓	✓	✓	✓	✓
Survey weights	✓	✓	✓	✓	✓	✓	✓

289 respondents, 23 tracts; side-specific distance gradients; robust SEs; sharpened q-values in brackets

On the east side, trust in neighbors goes with less schooling

Institutional trust | Memberships | Falsification

Same pattern beyond Guadalajara, as in the motivating evidence:

World Values Survey | SCI: regions | SCI: U.S. ZIP codes

Parents invest if the return beats the exchange put at risk

Parents invest in a child's education if

$$\underbrace{\rho - C_i}_{\text{net return; same } \rho \text{ for all}} \geq \underbrace{a(s)}_{\text{salience}} \underbrace{\Pi_i}_{\text{exchange put at risk}}$$

C_i: school provision, parental schooling, household resources

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C_i: school provision, parental schooling, household resources

Π_i is larger when

- exchange with neighbors is worth more: worse outside options, more neighbors taking part
- the household takes part and uses the network more
- the child is an adolescent: more time taken, closer to leaving

Parents invest if the return beats the exchange put at risk

Parents invest in a child's education if

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Map-First raises $a(s)$: demand falls only where $\Pi_i > 0$
the experiment varies salience, not Π_i

Full static model Dynamics

Experiment: map the neighborhood before or after an educational choice

- 239 households
  - parent or caregiver; child aged 6–20
  - 193 east, 46 west; within 3 km

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- **Choice**: 1 hour of math tutoring vs. an electronic gift card
  - price list (Jack et al., 2022): 11 rows, MXN 0–250
  - one row binds; 1 in 10 paid

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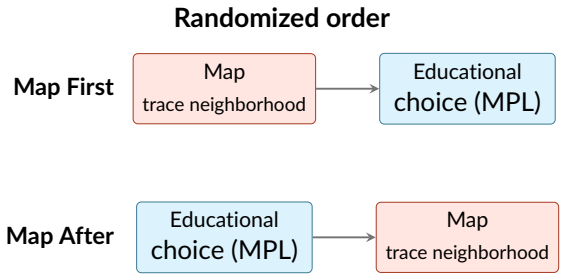
- **Treatment**: map the neighborhood before or after the choice
  - trace neighborhood: *barrio* or *colonia*
  - no information or incentives: **only salience**



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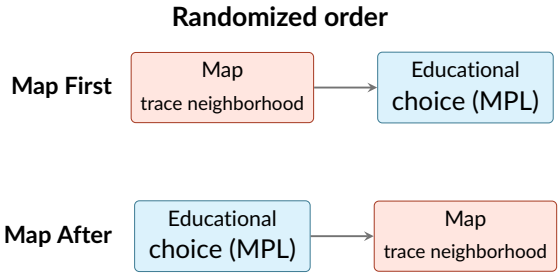
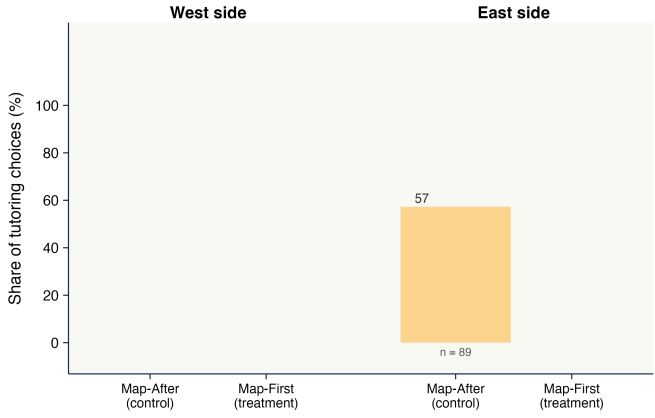


Pre-registered: AEARCTR-0018600

- **H1**: Map-First lowers demand for tutoring
- **H2**: the effect is concentrated east of the boundary

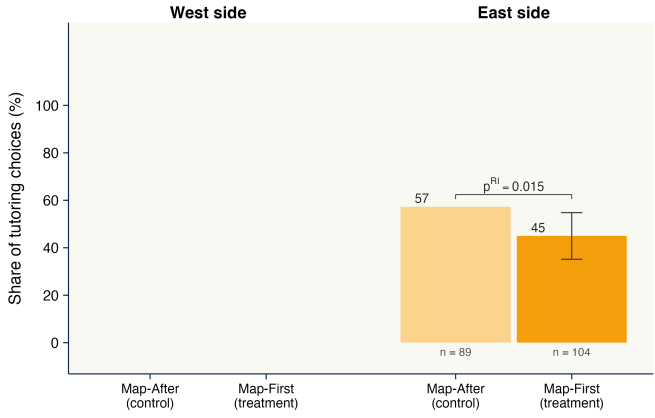
Choice list | Balance | Comprehension

Neighborhood salience lowers tutoring demand on the east side



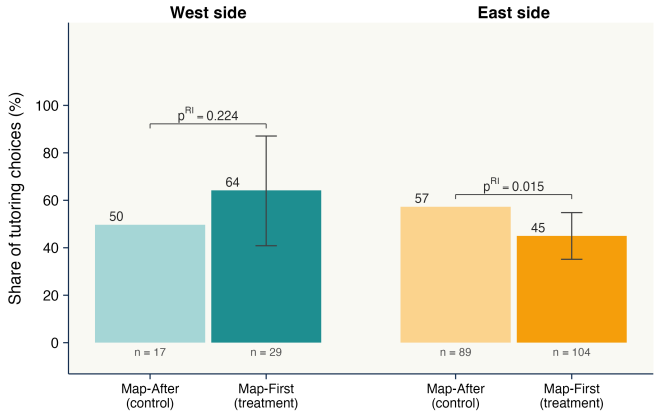
East, Map-After: tutoring chosen in 57% of the 11 choices

Neighborhood salience lowers tutoring demand on the east side



East: Map-First lowers the tutoring share by 12.3 pp, about one fifth

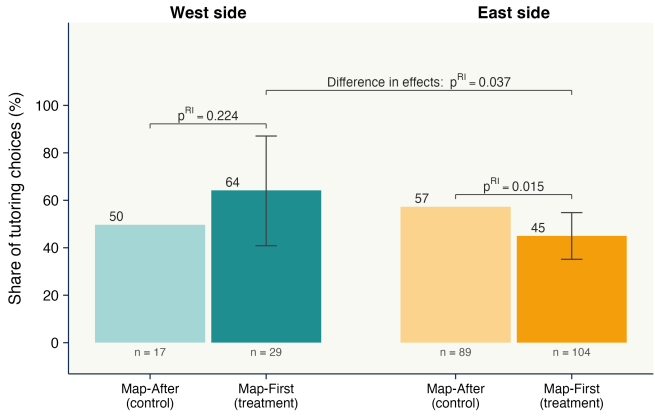
Neighborhood salience lowers tutoring demand on the east side



West: positive and imprecise, with 46 households

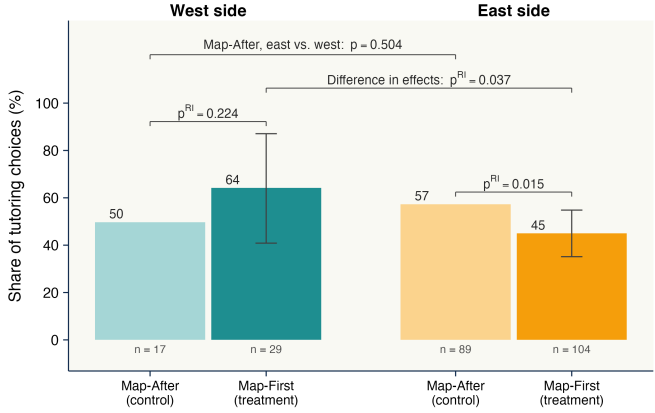
Map-First households there: 10.6 fewer years in the neighborhood

Neighborhood salience lowers tutoring demand on the east side



The effect differs between the two sides

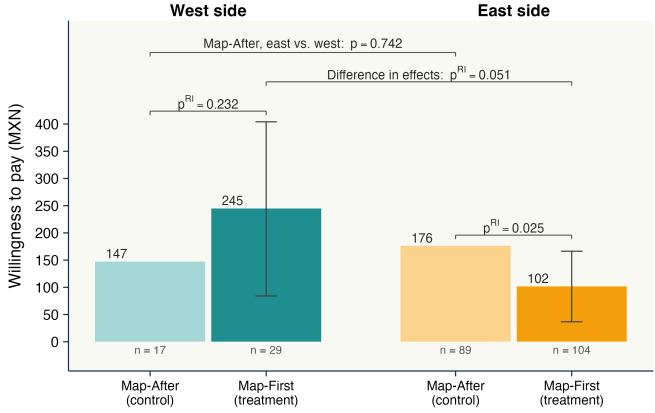
Neighborhood salience lowers tutoring demand on the east side



Same demand without the prime:
not a different preference for education, but a different trade-off with the neighborhood

Table | Balance | Diagnostics | Leave-one-out | Bounded WTP | Experimenter demand | Heterogeneity | What the map makes salient | Sample size

Neighborhood salience lowers tutoring demand on the east side

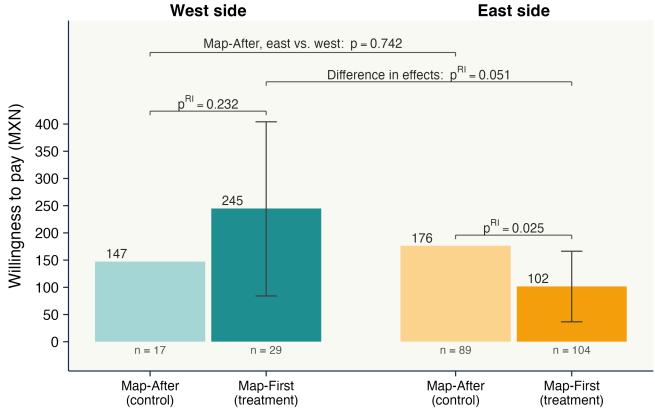


Willingness to pay falls by MXN 75 in the east, about two fifths

about a quarter of a daily minimum wage

Table | Balance | Diagnostics | Leave-one-out | Bounded WTP | Experimenter demand | Heterogeneity | What the map makes salient | Sample size

Neighborhood salience lowers tutoring demand on the east side



0.33 and 0.30 SD: close to the average priming effect on behavior ($d = 0.38$)

above bias-corrected estimates of 0.20–0.28 (Dai et al., 2023); here pre-registered and incentivized

Table | Balance | Diagnostics | Leave-one-out | Bounded WTP | Experimenter demand | Heterogeneity | What the map makes salient | Sample size

Interpretation: what lies behind the east–west difference

Survey
experiment
households

East: more exchange with neighbors, same valuation of education

help twice as often; same aspirations and returns

Exchange Beliefs Norms SES

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Prime
the historical divide

East: money explains dropout less often
money or work: −15 pp

Explanations Table

Interpretation: what lies behind the east–west difference

Survey experiment households	East: more exchange with neighbors, same valuation of education help twice as often; same aspirations and returns	Exchange · Beliefs · Norms · SES
Prime the historical divide	East: money explains dropout less often money or work: −15 pp	Explanations · Table
1821 civil census	Social life already locally organized sons named after their parish saint	Names

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Exchange with neighbors, not a lower valuation of education

Magnitudes: Consistency check

Conclusion

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In a place shaped by historical exclusion,
does the neighborhood affect the demand for education?

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- **Persistence**: a schooling gap at the colonial boundary outlives material convergence

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Where exclusion turned social life inward, the neighborhood competes with education

Thanks!

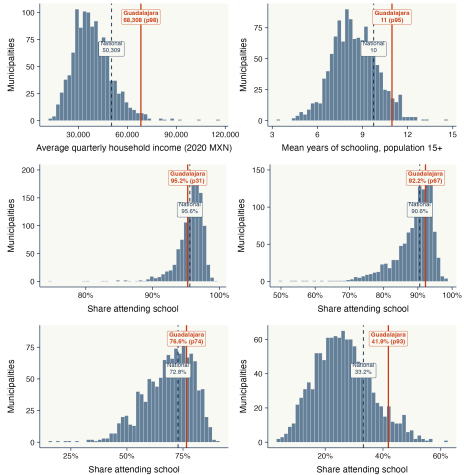
guillermo.woo-mora@psemail.eu

Appendix

Appendix: Context

Guadalajara is rich and schooled by national standards

1,169 urban municipalities, 2020 Census; solid: Guadalajara (value, percentile); dashed: national

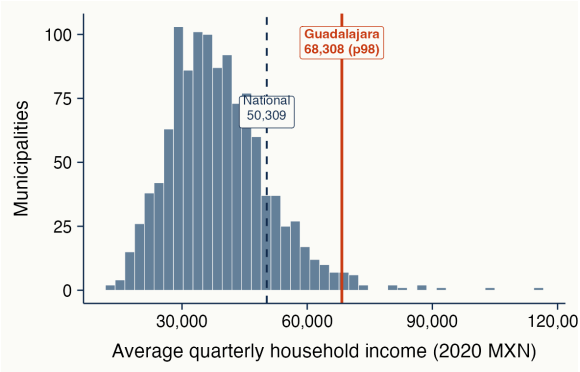


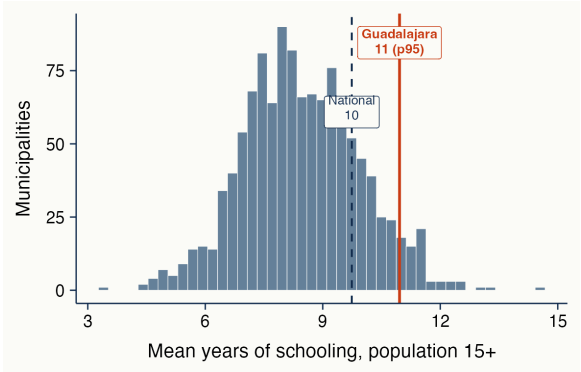
income; schooling (15+); attendance at ages 6–11, 12–14, 15–17, 18–24

Back: Guadalajara | Income and schooling

Guadalajara is at the 98th percentile of income and the 95th of schooling

1,169 urban municipalities, 2020 Census

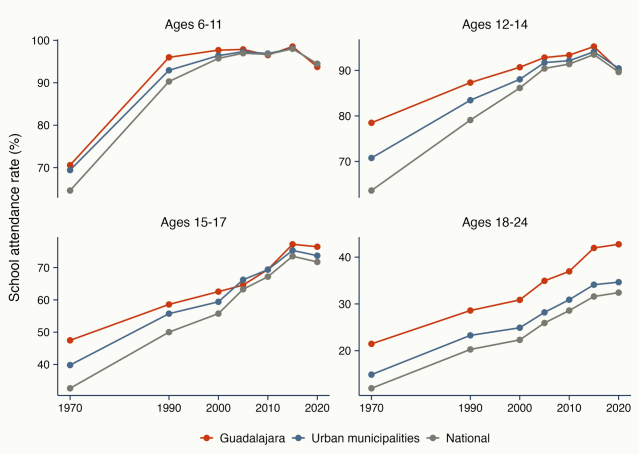




Back: Guadalajara National distribution

Attendance beyond primary has stayed above the national rate

IPUMS International census samples; urban municipalities: $\geq$50% urban population, excluding Guadalajara

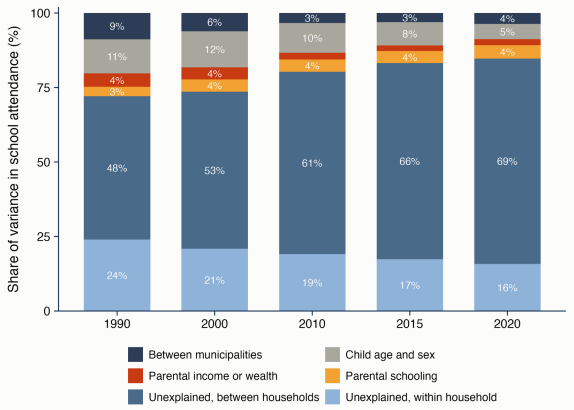


Ages 18–24: 21.5% vs. 12.0% nationally in 1970; 42.7% vs. 32.4% in 2020

Back: Guadalajara

Parental resources explain little of the variance in attendance

Ages 12–17; within-municipality variance split with municipality FE, Shapley values over all orderings
income: head and spouse labor earnings per member; wealth: assets and dwelling index; schooling: more schooled of head and spouse

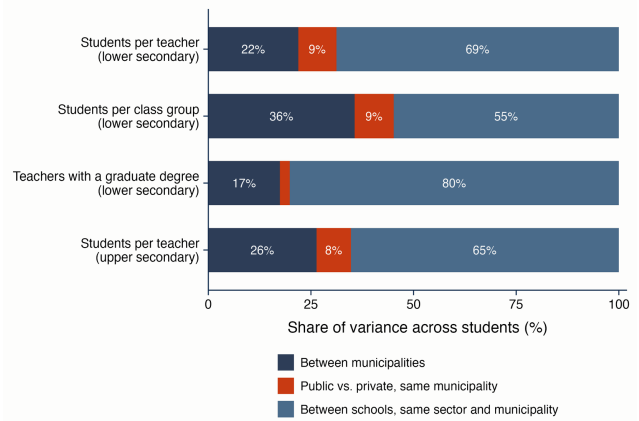


2020: parental schooling 4.5%, income or wealth 2.0%, between municipalities 3.7%

Back: Guadalajara

School supply varies mostly within municipalities

Formato 911, 2018–19, national; variance across students, schools weighted by enrollment

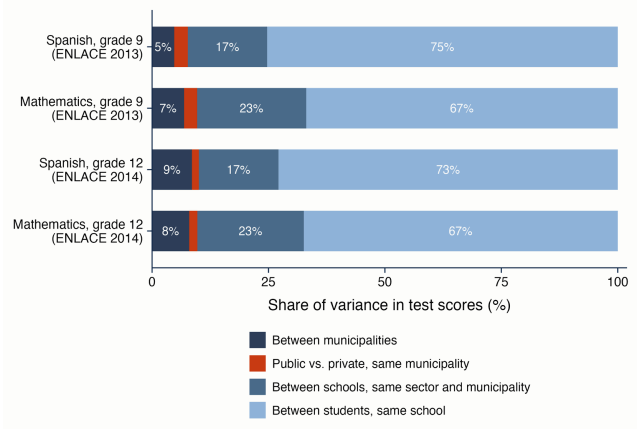


attending a private school is a household choice; between schools combines supply with sorting of students

Back: Guadalajara

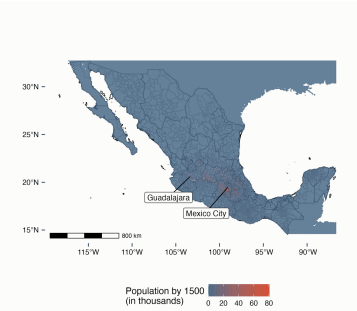
Test scores vary mostly between students of the same school

ENLACE individual scores, national; grade 9 (2013) without students flagged for copying; grade 12 (2014) without incomplete tests

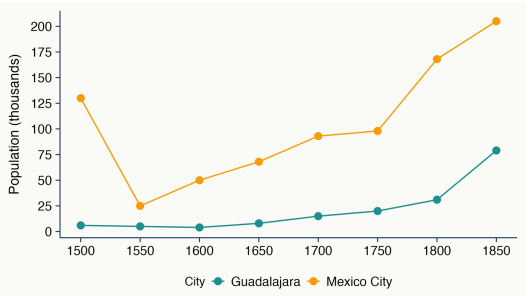


Back: Guadalajara

The region had no urban centers before the conquest



Population density around 1500 (Goldewijk et al., 2017)



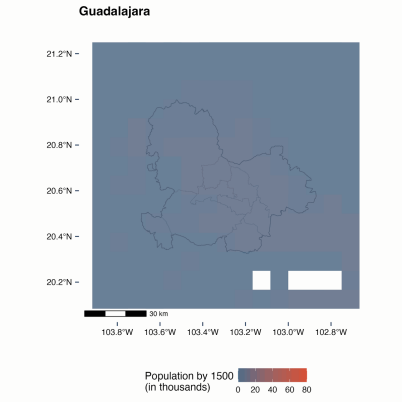
Urban population, 1500–1850 (Buringh, 2018)

Northern pre-Hispanic territories were "a congeries of stateless or, more dramatically, acephalic societies" (Knight, 2002)

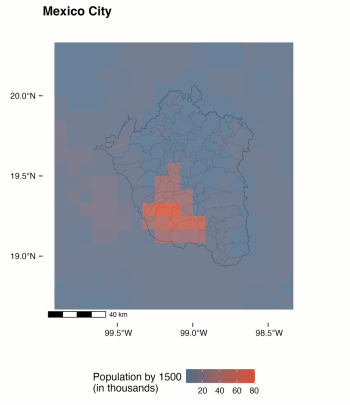
Back: Setting

The region had no urban centers before the conquest

Goldewijk et al. (2017) historical population estimates



Atemajac Valley (Guadalajara)

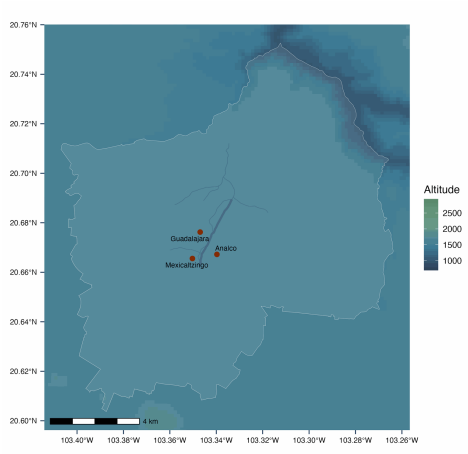


Anahuac Valley (Mexico City)

Northern pre-Hispanic territories were "a congeries of stateless or, more dramatically, acephalic societies" (Knight, 2002)

Back: Setting

Terrain and climate do not jump at the creek



Altitude, 250 × 250 m cells

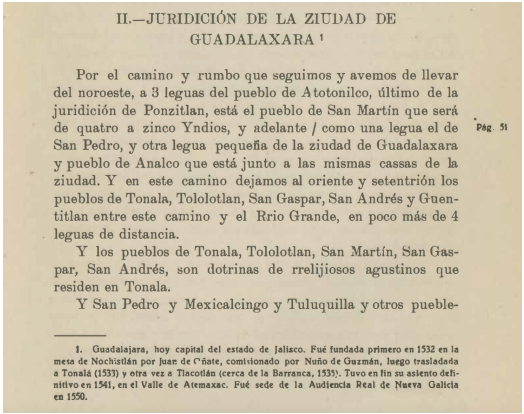
RDD at the grid-cell level, 1.5 km buffer; standardized; robust bias-corrected CIs, Calonico et al. (2014) bandwidths; sharpened q-values

	RDD estimate	SE	95% CI	p-value	FDR q-value
Altitude	0.002	0.015	[−0.027, 0.031]	0.910	0.922
Slope	-0.053	0.039	[−0.128, 0.023]	0.174	0.646
Temperature	-0.004	0.038	[−0.078, 0.070]	0.922	0.922
Precipitation	-0.163	0.144	[−0.446, 0.120]	0.258	0.646
Humidity	-0.036	0.055	[−0.143, 0.071]	0.510	0.766
Thermal variation	0.154	0.156	[−0.152, 0.460]	0.323	0.646

Altitude and slope from a 15 m elevation model: precise nulls

Back: Setting

Both settlements were governed by the city council



II.—JURIDICIÓN DE LA ZIUDAD DE GUADALAXARA [1]

Por el camino y rumbo que seguimos y avemos de llevar del noroeste, a 3 leguas del pueblo de Atotonilco, último de la juridición de Ponzitlan, está el pueblo de San Martín que será de quatro a zinco Yndios, y adelante / como una legua el de San Pedro, y otra legua pequeña de la ziudad de Guadalaxara y pueblo de Analco que está junto a las mismas cassas de la ziudad. Y en este camino dejamos al oriente y setentrión los pueblos de Tonala, Tololotlan, San Gaspar, San Andrés y Guentitlan entre este camino y el Rrio Grande, en poco más de 4 leguas de distancia.

Pág. 51

Y los pueblos de Tonala, Tololotlan, San Martín, San Gaspar, San Andrés, son dotrinas de rrelijiosos agustinos que residen en Tonala.

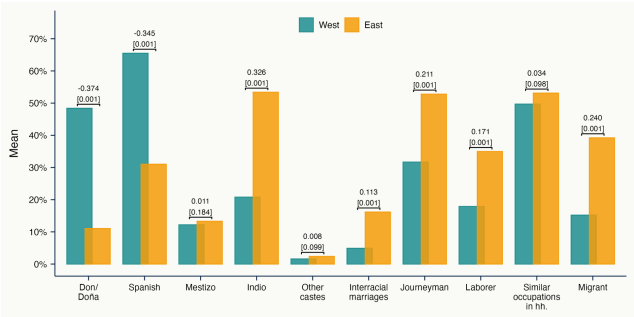
Y San Pedro y Mexicalcingo y Tuluquilla y otros pueble-

1. Guadalajara, hoy capital del estado de Jalisco. Fué fundada primero en 1532 en la mesa de Nochistlán por Juan de Oñate, comisionado por Nuño de Guzmán, luego trasladada a Tonalá (1533) y otra vez a Tlacotlán (cerca de la Barranca, 1535). Tuvo en fin su asiento definitivo en 1541, en el Valle de Atemaxac. Fué sede de la Audiencia Real de Nueva Galicia en 1550.

- Both banks under the colonial authorities of Guadalajara and the Audiencia of New Galicia (Lázaro de Arregui, 1946; Solís Matías, 1992)
- For most of the colonial period, neither Analco nor Mexicaltzingo held the autonomy of a *República de Indios*
- Around 1800 both are absent from the *propios y arbitrios* accounts: under the city's *ayuntamiento* (Tanck de Estrada, 1999)

Back: Setting

In 1821, both banks were mixed, but the east ranked lower



Means by bank, 1821 census (Anderson and Spike, 2007)

	Full sample	*Calidad* subsample	*Calidad* controlled
Don/Doña	-0.374***	-0.337***	-0.176***
	(0.016)	(0.021)	(0.022)
Interracial marriages	0.113***	0.113***	0.103***
	(0.018)	(0.018)	(0.018)
Journeyman	0.211***	0.195***	0.130***
	(0.019)	(0.027)	(0.029)
Laborer	0.171***	0.129***	0.089***
	(0.017)	(0.025)	(0.026)
Similar occupations in hh.	0.034	-0.054	-0.071*
	(0.029)	(0.040)	(0.043)
Migrant	0.240***	0.073***	0.088***
	(0.016)	(0.026)	(0.027)

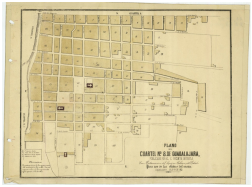
East minus west, with and without a control for *calidad*

Sixteen *cuarteles* adjacent to the creek; sex and age controls; standard errors clustered by household in parentheses; * and *** mark the 10% and 1% levels

Back: Setting

Four city maps, 1800–1908, locate named institutions

1800



1863, *cuartel* 8 (Analco)

1890

1908

Category	1800 West	1800 East	1863 West	1863 East	1890 West	1890 East	1908 West	1908 East
Educational	4	0	3	0	10	0	14	1
Church	15	3	18	3	0	0	21	3
Government	5	0	5	0	12	1	5	0
All twelve categories	28	3	49	7	76	7	54	8

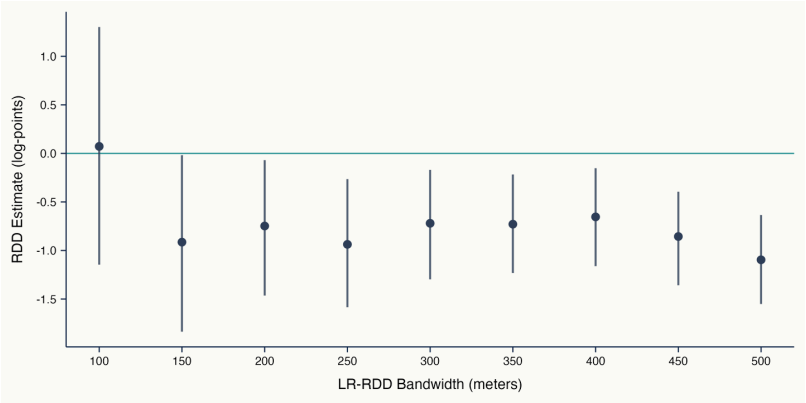
Located entries by category, map and bank

The 1890 index has no separate church section

Back: Material exclusion

Land was worth half to two-thirds less on the east bank in 1882

1882 cadaster, log assessed land value per m^2 by block; local-randomization RDD at the creek (Cattaneo et al., 2015)



In windows of 150 to 500 m, assessed land values are 48 to 67 percent lower on the east bank

Back: Material exclusion

Spatial RDD: school supply and quality do not differ at the boundary

Block-level spatial RDD, 13,791 census blocks of Guadalajara; schools from the SEP directory 2018–2019

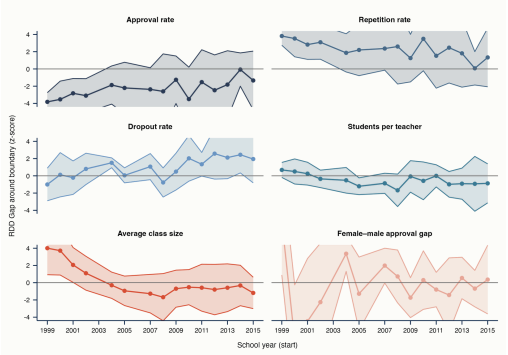
	Mean within h							
Outcome	West	East	$\hat{\tau}$	SE	p-val	q-val	h (m)	N_h
Total schools	0.138	0.127	0.018	0.063	0.713	1.000	2,068	2,739
Total schools $\times$ shifts	0.148	0.137	0.011	0.072	0.848	1.000	2,088	2,771
Public schools	0.055	0.064	0.015	0.041	0.742	1.000	2,206	3,004
Private schools	0.097	0.071	-0.007	0.056	0.937	1.000	1,891	2,368
Primary schools	0.035	0.045	0.005	0.024	0.899	1.000	2,208	3,015
Secondary schools	0.013	0.017	-0.005	0.012	0.465	1.000	1,882	2,349
High schools	0.027	0.022	-0.042	0.032	0.110	1.000	3,337	5,637
Public share	0.268	0.475	0.264	0.239	0.296	1.000	1,384	112
Students per teacher	16.2	17.6	0.218	4.457	0.938	1.000	2,308	227

Notes: Each row is a separate spatial RDD, east minus west, with MSE-optimal bandwidth h, side-specific linear fits, longitude, latitude and boundary-segment fixed effects; robust bias-corrected SE and p-value (Calonico et al., 2014). West/East: mean outcome within h of the boundary on each side. N_h: observations within h. q: sharpened FDR q-values (Anderson, 2008); rows with $q \leq 0.10$ shaded. Outcomes in levels: counts of schools per block (blocks without a school enter as zero); the public share and students per teacher are defined only for blocks with a school. q across the nine outcomes; none is shaded.

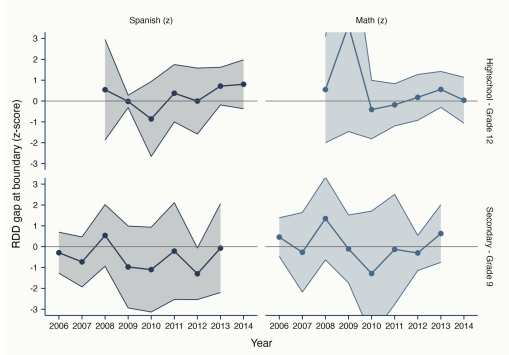
Back: Local conditions

Spatial RDD: school supply and quality do not differ at the boundary

School quality, spatial RDD by year for schools within 1.5 km of the boundary; 95% bands



Secondary schools, Formato 911, 1998–2015



ENLACE test scores, 2006–2014

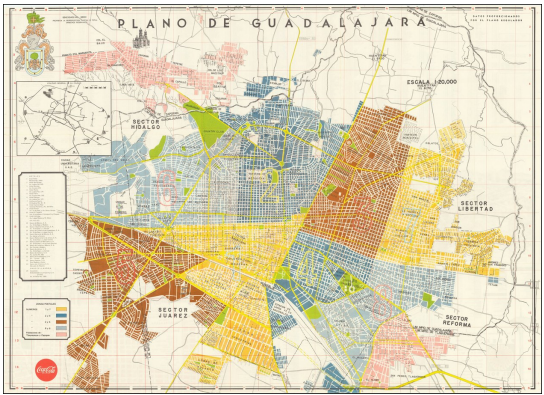
The east lagged in approval, repetition and class size until about 2002; no gap since the mid-2000s

Back: Local conditions

The creek became an avenue that connects the two banks

San Juan de Dios creek → Calzada Independencia

Guadalajara, 1908

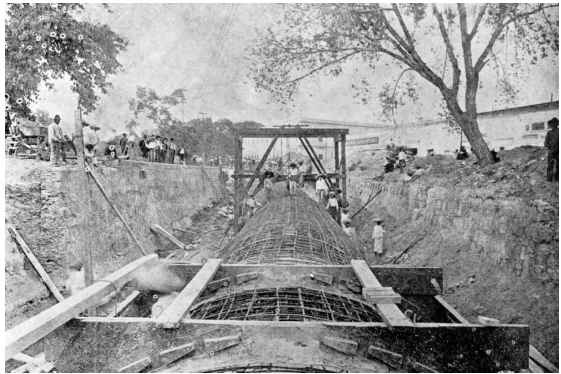


Aerial view, 1972

Back: The boundary today

The creek became an avenue that connects the two banks

San Juan de Dios creek → Calzada Independencia



Piping the creek, 1903–1911

Back: The boundary today

The creek became an avenue that connects the two banks

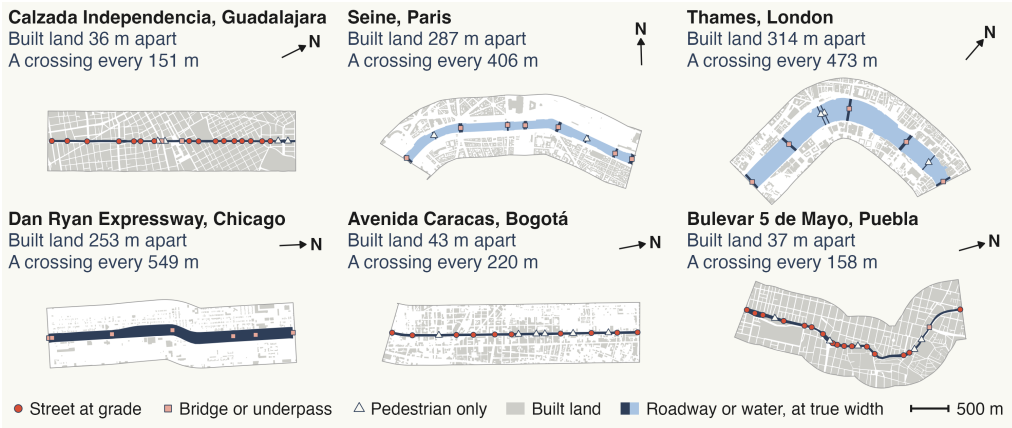
San Juan de Dios creek → Calzada Independencia

Looking west, 2020

Looking east, 2020

Back: The boundary today

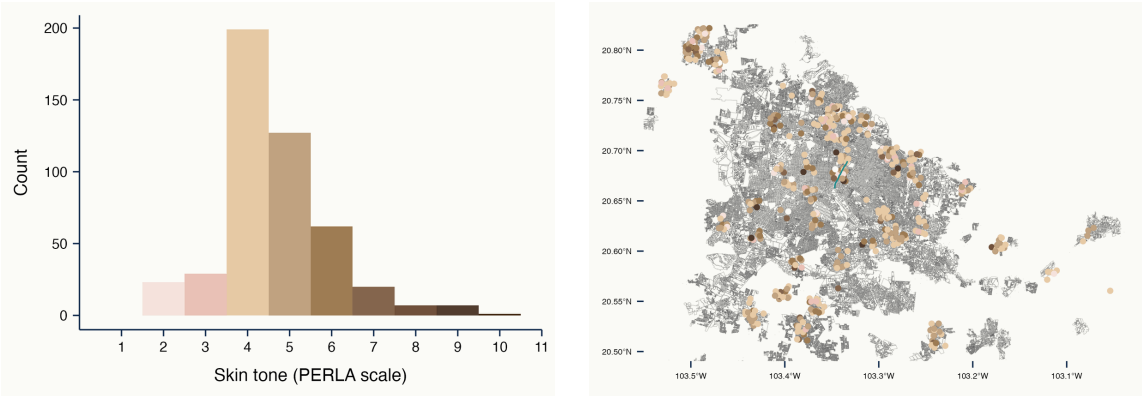
The avenue is crossed almost three times as often as the Seine



Stretches of about 3.3 km, same scale. Seine: Pont du Carrousel to Pont d'Iéna. Thames: Lambeth to Southwark Bridge. Chicago: Pershing Road to Garfield Boulevard. Bogotá: Calle 32 to Calle 63. Puebla: 32 Poniente to 27 Oriente.

Back: The boundary today

Skin tone is not clustered by side of the boundary

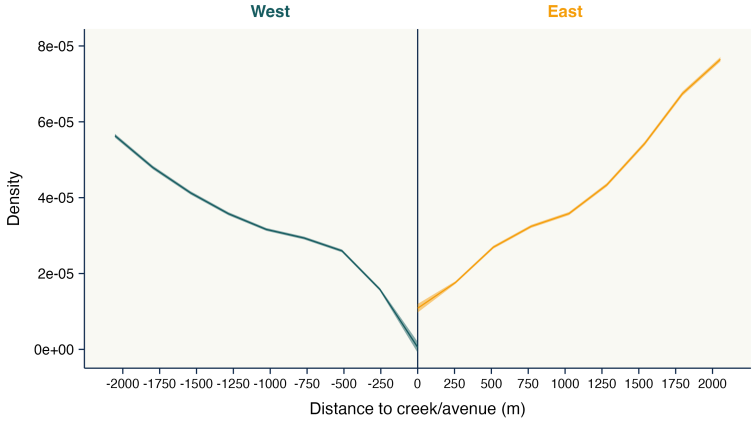


Interviewer-rated skin tone, 2017 mobility survey

Back: The boundary today

Appendix: Persistent Human Capital Gaps

Density is higher east of the boundary, with no spike at the line



Density higher east at the cutoff; no spike at the line (Cattaneo et al., 2018)

West edge: the historic commercial core, where few people live

Back: Design

The gap is negative in every specification, less precisely for attendance

2020 Census; preferred: third row of Panel A, fourth row of Panel B

$\hat{\tau}$	SE	95% CI	p-val	Poly	h	N_h	b	N_b	Kernel	VCE	Clean ctrls	Unclean ctrls	Specification
Panel A. Years of schooling (ages 6+)													
-0.791	0.156	[-1.097, -0.485]	0.000	1	570.750	25011	1489.073	107104	Uniform	NN	No	No	Main – levels
-0.165	0.033	[-0.229, -0.101]	0.000	1	570.750	25011	1489.073	107104	Uniform	NN	No	No	Main – z-score
-0.141	0.048	[-0.235, -0.047]	0.003	1	395.465	13833	769.946	42070	Uniform	NN	✓	No	Main – z-score + covariates
-0.117	0.041	[-0.198, -0.036]	0.005	1	410.825	13465	796.536	40669	Uniform	NN	✓	✓	Main – z-score + covariates + clean controls
-0.151	0.047	[-0.244, -0.058]	0.001	2	741.235	39262	1461.118	104303	Uniform	NN	✓	No	Rob. – Polynomial order (2)
-0.103	0.047	[-0.194, -0.012]	0.027	1	477.508	18525	897.343	51626	Triangular	NN	✓	No	Rob. – Kernel (triangular)
-0.127	0.044	[-0.212, -0.042]	0.004	1	504.318	20665	913.445	53039	Epanechnikov	NN	✓	No	Rob. – Kernel (Epanechnikov)
-0.081	0.051	[-0.181, 0.018]	0.110	1	388.115	13627	759.784	41158	Uniform	HC3	✓	No	Rob. – VCE (HC3, most conservative)
-0.122	0.043	[-0.205, -0.039]	0.004	1	407.383	12717	796.109	38784	Uniform	NN	✓	✓	Rob. – Migrants (native-born only)
Panel B. School attendance (ages 6–22)													
-0.032	0.019	[-0.070, 0.005]	0.094	1	1200.216	18265	2058.598	45016	Uniform	NN	No	No	Main – levels
-0.075	0.045	[-0.163, 0.013]	0.094	1	1200.216	18265	2058.598	45016	Uniform	NN	No	No	Main – z-score
-0.104	0.072	[-0.245, 0.036]	0.146	1	571.045	6101	1081.865	15789	Uniform	NN	✓	No	Main – z-score + covariates
-0.086	0.047	[-0.178, 0.006]	0.067	1	894.781	12172	1997.570	41693	Uniform	NN	✓	✓	Main – z-score + covariates + clean controls
-0.129	0.082	[-0.289, 0.032]	0.116	2	871.024	12038	1800.565	35228	Uniform	NN	✓	No	Rob. – Polynomial order (2)
-0.136	0.063	[-0.260, -0.011]	0.032	1	703.139	8847	1618.948	29378	Triangular	NN	✓	No	Rob. – Kernel (triangular)
-0.124	0.068	[-0.257, 0.009]	0.067	1	616.138	7038	1472.523	24574	Epanechnikov	NN	✓	No	Rob. – Kernel (Epanechnikov)
-0.091	0.078	[-0.244, 0.063]	0.246	1	504.945	5117	1028.982	14819	Uniform	HC3	✓	No	Rob. – VCE (HC3, most conservative)

Robust bias-corrected estimates (Calonico et al., 2014). Clean controls: sex, age, longitude, latitude, boundary-segment FE. Unclean, Panel A: Afro-descendant or Indigenous self-identification, born in Jalisco, residence five years earlier, social security, marital status, dwelling type. Unclean, Panel B: parental schooling, household size, residence five years earlier. Levels rows in original units; all others standardized.

Back: Years of schooling

Characteristics that differ at the boundary enter the estimates as controls

2020 Census, ages 6+: a spatial RDD on each characteristic, east minus west, without covariates

Characteristic	At the boundary West	At the boundary East	$\hat{\tau}$	SE	p-val	q-val
Panel A. Demographics, origin and ethnicity						
Age (years)	39.81	39.05	-0.76	0.69	0.268	0.241
Male	0.493	0.498	0.005	0.010	0.598	0.369
Catholic	0.877	0.881	0.004	0.011	0.742	0.467
Born in Jalisco	0.812	0.861	0.049	0.013	0.000	0.001
In Guadalajara 5 years earlier	0.957	0.976	0.019	0.005	0.000	0.001
Speaks an Indigenous language	0.004	0.014	0.011	0.002	0.000	0.001
Panel B. Health insurance						
None	0.298	0.280	-0.018	0.012	0.130	0.150
Public	0.641	0.659	0.018	0.011	0.106	0.130
Private	0.025	0.012	-0.013	0.004	0.003	0.007
Other institution	0.004	0.064	0.060	0.007	0.000	0.001
Panel C. Marital status						
Single	0.456	0.450	-0.006	0.012	0.594	0.369
Married	0.243	0.242	-0.001	0.008	0.939	0.624
Cohabiting	0.120	0.131	0.011	0.010	0.281	0.241
Separated or divorced	0.081	0.084	0.003	0.004	0.417	0.284
Widowed	0.062	0.065	0.003	0.003	0.289	0.241
Panel D. Activity						
Works	0.600	0.606	0.006	0.008	0.451	0.292
Unemployed	0.063	0.048	-0.015	0.006	0.008	0.015
Housework	0.082	0.096	0.014	0.006	0.022	0.036
Retired	0.064	0.042	-0.022	0.006	0.000	0.001
Unable to work	0.006	0.003	-0.003	0.002	0.069	0.089

With these characteristics as controls, the schooling gap is -0.117 SD, against -0.141 SD without them

Notes: Each row is a separate spatial RDD on individuals aged 6 and older, with a uniform kernel and MSE-optimal bandwidth; robust bias-corrected estimate, SE and p-value (Calonico et al., 2014). West, East: fitted values at the boundary on each side. q: sharpened FDR q-values across the 22 characteristics estimated, of which 20 are shown (Anderson, 2008); rows with $q \leq 0.10$ shaded. The specification with all controls adds Afro-descendant and Indigenous-language status, birthplace, residence five years earlier, health insurance, marital status, activity and dwelling type.

All specifications · Back: Years of schooling

Excluding inter-state migrants leaves the gap unchanged

- 91.5% of residents lived in Jalisco five years earlier
- Excluding inter-state migrants: -0.12 SD vs. -0.14 SD

Back: Years of schooling

Excluding inter-state migrants leaves the gap unchanged

- 91.5% of residents lived in Jalisco five years earlier
- Excluding inter-state migrants: -0.12 SD vs. -0.14 SD
- Long-term residence proxy: 2020 adults (34+) matched to similar 2010 residents of the same block; controls: sex, age, longitude, latitude, boundary-segment FE

Outcome	$\hat{\tau}$	SE	95% CI	p-val	Poly	h	b	N_h	N_b
Probability to stay in place	0.144	0.075	[-0.003, 0.291]	0.054	1	131.9	254.9	1,315	3,672
Probability to stay in place (with controls)	0.178	0.080	[0.022, 0.334]	0.026	1	116.3	208.9	1,121	2,605

Back: Years of schooling

Excluding inter-state migrants leaves the gap unchanged

- 91.5% of residents lived in Jalisco five years earlier
- Excluding inter-state migrants: -0.12 SD vs. -0.14 SD
- Long-term residence proxy: 2020 adults (34+) matched to similar 2010 residents of the same block; controls: sex, age, longitude, latitude, boundary-segment FE

Outcome	$\hat{\tau}$	SE	95% CI	p-val	Poly	h	b	N_h	N_b
Probability to stay in place	0.144	0.075	[-0.003, 0.291]	0.054	1	131.9	254.9	1,315	3,672
Probability to stay in place (with controls)	0.178	0.080	[0.022, 0.334]	0.026	1	116.3	208.9	1,121	2,605

- Not a measured difference in turnover
  anonymous records; match distance falls mechanically with more records, and density is higher east
- Silent on who left: turnover volume is not selection

Back: Years of schooling

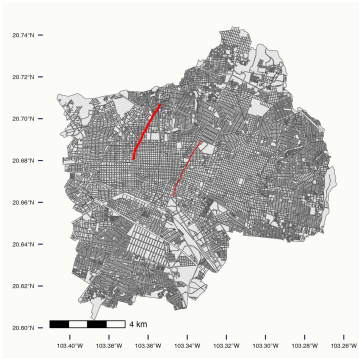
The gap remains when residents nearest the boundary are excluded

2020 Census, standardized units; excluding residents within 50, 100, 150 m; no covariates in any row
uniform kernel, local linear, NN variance; bandwidth re-selected for each radius

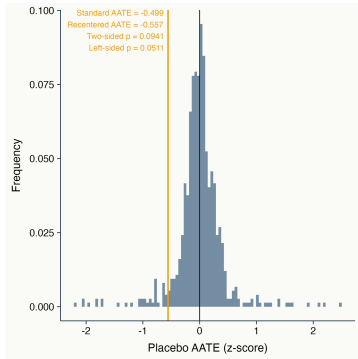
$\hat{\tau}$	SE	95% CI	p-val	h	N_h	Donut (m)
Panel A. Years of schooling (ages 6+)						
-0.165	0.033	[-0.229, -0.101]	0.000	570.8	25011	0
-0.101	0.026	[-0.151, -0.050]	0.000	976.0	57387	50
-0.211	0.031	[-0.272, -0.149]	0.000	667.2	32464	100
-0.151	0.031	[-0.212, -0.091]	0.000	916.7	49857	150
Panel B. School attendance (ages 6–22)						
-0.075	0.045	[-0.163, 0.013]	0.094	1200.2	18265	0
-0.086	0.047	[-0.178, 0.007]	0.069	1022.6	14606	50
-0.100	0.048	[-0.194, -0.006]	0.037	1098.3	15686	100
-0.127	0.056	[-0.237, -0.016]	0.025	954.9	12429	150

Back: Years of schooling

The true boundary lies in the tail of 1,000 placebo boundaries



One of 1,000 shifted and rotated copies of the boundary; 984 fall inside the municipality



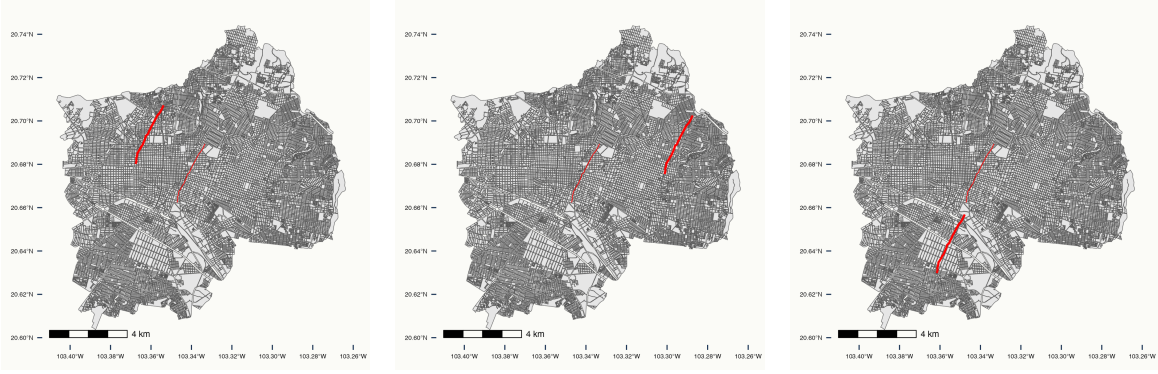
Block-level boundary-adaptive RDD, years of schooling

True boundary: -0.50 SD; recentered (Borusyak and Hull, 2023): -0.56 SD

Back: Years of schooling

The true boundary lies in the tail of 1,000 placebo boundaries

Three of the 1,000 shifted and rotated copies of the boundary



Back: Years of schooling

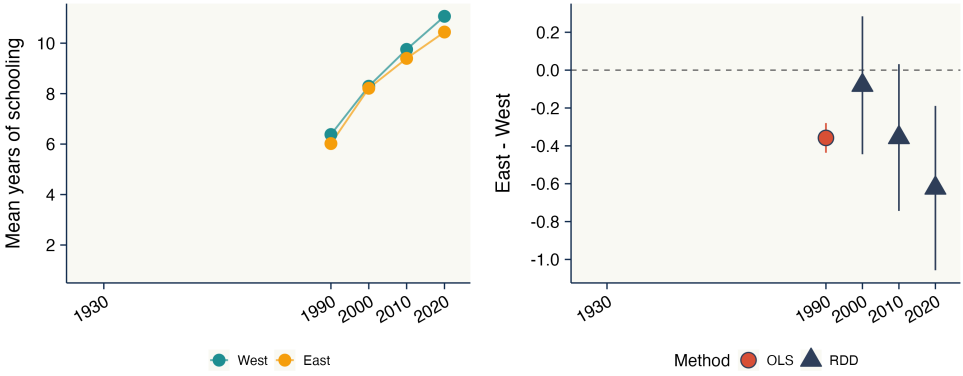
In 1930, east-side residents were less literate and less likely to be studying

Cuarteles adjacent to the boundary; sex and flexible age controls; SEs clustered by household

Variable	West Mean	East Mean	Diff.	Std. Error	*q*-value	*N*
Property: Some	0.233	0.212	-0.013	0.023	0.349	1,315
Property: Residence	0.023	0.020	-0.003	0.003	0.251	9,238
Birthplace: Jalisco	0.877	0.931	0.052	0.006	0.001	9,169
Occupation: Zapatero	0.009	0.025	0.015	0.003	0.001	9,172
Occupation: Merchant	0.050	0.052	0.002	0.004	0.390	9,172
Occupation: Laborer	0.032	0.035	0.002	0.004	0.349	9,172
Occupation: Mason	0.013	0.015	0.002	0.002	0.274	9,172
Occupation: Employed in firm	0.019	0.014	-0.005	0.003	0.082	9,172
Occupation: Maid	0.016	0.005	-0.011	0.002	0.001	9,172
Occupation: Housework	0.305	0.332	0.033	0.007	0.001	9,172
Human Capital: Illiterate	0.314	0.432	0.114	0.009	0.001	9,238
Human Capital: Studying	0.375	0.266	-0.106	0.012	0.001	3,965

Back: 1930

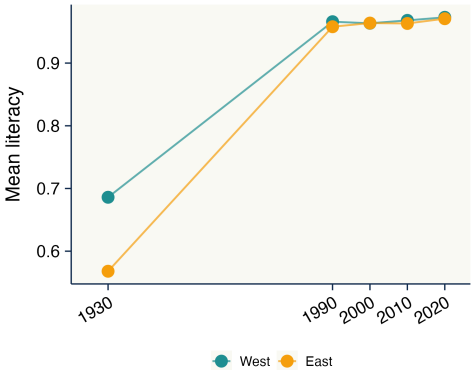
Years of schooling rose on both sides, and the gap remains in 2020

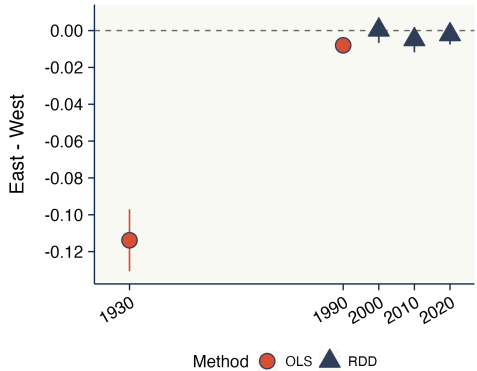


2000 estimate near zero: the 1992 explosions repopulated part of Analco inside the bandwidth

Back: 1930

Literacy converged





1930: 68.6% west vs. 56.8% east | from 1990: near-universal, no gap

Back: 1930

Household resources do not jump at the boundary

2015 Intercensal Survey, Guadalajara; spatial RDD at the historical boundary; survey weights

	Clean controls		+ dirty controls	
	$\hat{\tau}$ (SE)	N_h	$\hat{\tau}$ (SE)	N_h
A. Human capital				
Years of schooling	-0.305^{***} (0.097)	3,077	-0.213^{**} (0.101)	2,481
Attends school in Guadalajara	-0.037 (0.032)	1,038		
B. Household resources				
Household income (log)	-0.281 (0.194)	832	-0.102 (0.201)	592
Government transfers	-0.013 (0.073)	867	$+0.009$ (0.063)	1,092
Pension income	-0.067 (0.092)	832	$+0.001$ (0.057)	1,632
International remittances	$+0.002$ (0.027)	2,745	-0.007 (0.035)	1,820
Domestic remittances	-0.005 (0.051)	1,551	-0.016 (0.046)	2,029

Conditioning on household income: schooling gap $-0.31 \rightarrow -0.21$ **SD**

Schooling in SD; income in log points; other rows are shares. Dirty controls: household income per capita (person level); also parental schooling, household head, distance to CBD (household level)

Back: Standard determinants | Credit and housing finance

Credit use today: no sign of a more constrained east side

2015 Intercensal Survey; Panels C–E: spatial RDD at the boundary; Panel F: municipality-wide comparison among workers, SEs clustered by dwelling

	Clean controls		+ dirty controls	
	$\hat{\tau}$ (SE)	N_h	$\hat{\tau}$ (SE)	N_h
C. Housing tenure				
Owner-occupied	+0.141 (0.102)	837	+0.247*** (0.068)	1,345
Rented	−0.170 (0.116)	561	−0.304*** (0.061)	2,129
Owned by relatives, or lent	−0.098 (0.067)	786	−0.056 (0.055)	1,319
Title in the respondent's name	+0.002 (0.060)	2,036	+0.158 (0.124)	512
D. How the dwelling was acquired				
Bought ready-built	+0.271* (0.148)	423	+0.503*** (0.187)	423
Had it built	−0.044 (0.085)	517	−0.040 (0.069)	603
Built by the household	−0.035 (0.077)	470	−0.039 (0.093)	342
Inherited	−0.253* (0.134)	602	−0.401** (0.157)	427
E. How the dwelling was financed				
Own resources	+0.143* (0.082)	686	+0.169*** (0.050)	2,136
Loan from family or friends	+0.055* (0.033)	1,359	+0.021 (0.019)	2,365
Bank or other institutional credit	−0.014 (0.027)	354	+0.086*** (0.023)	1,151
Public housing credit	+0.007 (0.025)	533	−0.011 (0.025)	1,161
Fully paid	+0.760** (0.331)	111	+0.257 (0.197)	154
Still paying	−0.760** (0.331)	111	−0.258 (0.197)	154
F. Employment-linked credit (workers; municipality-wide comparison, not an RDD)				
Housing credit through employment	−0.023 (0.021)			
Retirement savings account (SAR/AFORE)	−0.008 (0.020)			

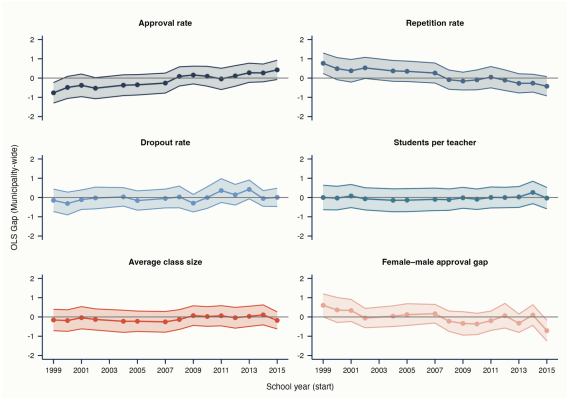
East: more owner-occupied and bought with own resources; no less bank, public, or employment-linked credit

Credit use today, not wealth, liquidity, collateral, or constraints when the schooling stock was formed

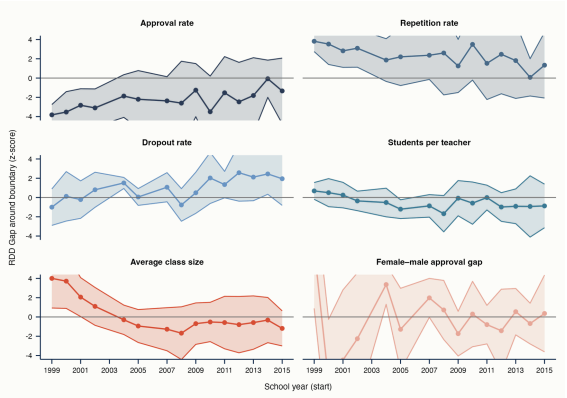
Back: Standard determinants | Household resources

School quality converged by the mid-2000s

Secondary schools, Formato 911, 1999–2015: east minus west, in standard deviations; 95% bands



Municipality-wide OLS

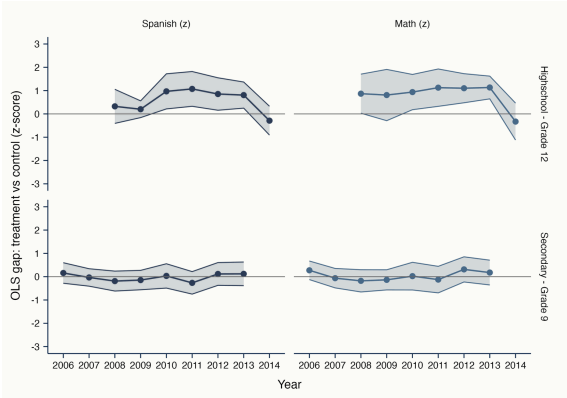


Spatial RDD, schools within 1.5 km of the boundary

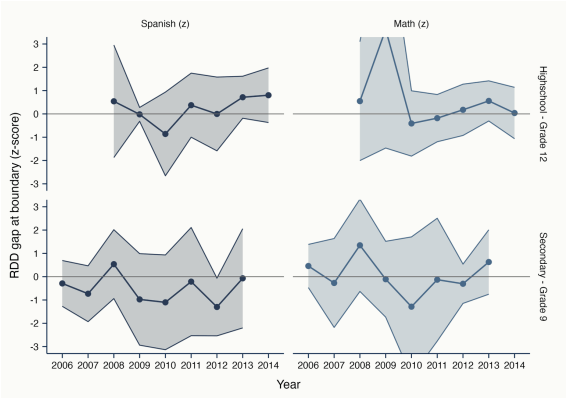
Definitions Test scores Back: Explanations

Test scores do not differ systematically across sides

ENLACE standardized tests, Spanish and math, grades 9 and 12, 2006–2014: east minus west, in standard deviations; 95% bands



Municipality-wide OLS



Spatial RDD at the boundary

School records (F911) Back: Explanations

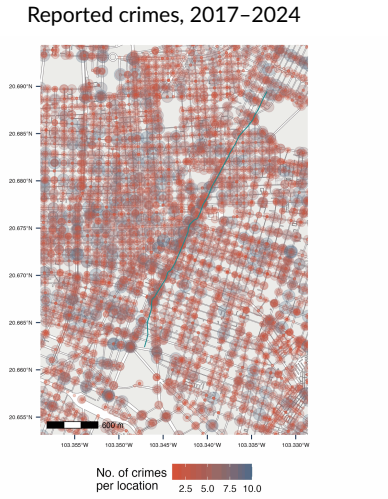
School quality: what each measure is

Measure	Definition (school-year level)	Source, years
Approval rate	students promoted at year end / students enrolled at year end (%)	*Formato 911*, secondary schools, 1998–2015
Repetition rate	repeaters / students enrolled at year end (%)	
Dropout rate	(initial enrollment − end-of-year enrollment) / initial enrollment (%)	
Average class size	students enrolled at year end / number of groups	
Students per teacher	students enrolled at year end / teachers	
Spanish, Math (z)	school mean ENLACE score, standardized within year and grade; grade 9 (secondary) and grade 12 (high school)	ENLACE, 2006–2014

Formato 911 rates are standardized within year; rates outside 0–100 and ratios above 100 are set to missing. OLS: municipality-wide, by year, with a public/private indicator and school coordinates, SEs clustered by school. RDD: schools within 1.5 km of the boundary, by year (Calonico et al., 2014).

Figures Back: Explanations

Crime exposure does not jump at the boundary



Back: Explanations

Crime exposure does not jump at the boundary

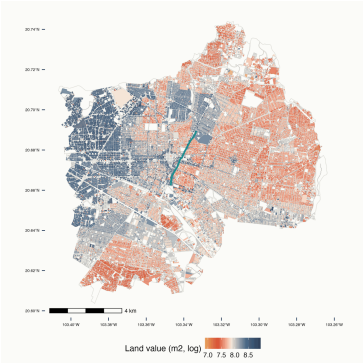
Block-level spatial RDD, 187,389 reports from the Jalisco Attorney General (IIEG) 2017–2024 geocoded to census blocks

	Mean within h							
Outcome	West	East	$\hat{\tau}$	SE	p-val	q-val	h (m)	N_h
Panel A. Total crime								
All crimes (log density per km^2)	0.701	0.766	0.210	0.247	0.514	1.000	748	702
All crimes (log rate per 1,000 residents)	0.645	0.501	0.099	0.239	0.791	1.000	813	789
Panel B. By crime type								
Property crimes (log density per km^2)	0.629	0.582	0.099	0.200	0.778	1.000	725	673
Property crimes (log rate per 1,000 residents)	0.581	0.372	-0.002	0.203	0.873	1.000	800	769
Violent crimes (log density per km^2)	0.131	0.141	0.045	0.072	0.575	1.000	828	806
Violent crimes (log rate per 1,000 residents)	0.116	0.095	0.020	0.079	0.824	1.000	866	850
Homicide (log density per km^2)	0.012	0.039	0.021	0.018	0.369	1.000	781	745
Homicide (log rate per 1,000 residents)	0.012	0.019	0.023	0.016	0.169	1.000	1,002	1,030
Violence against women (log density per km^2)	0.141	0.227	0.175	0.086	0.044	0.793	849	834
Violence against women (log rate per 1,000 residents)	0.106	0.145	0.117	0.073	0.103	0.861	824	803

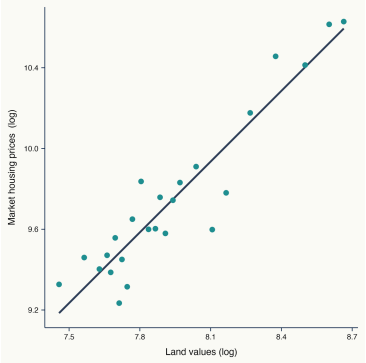
Notes: Each row is a separate spatial RDD, east minus west, with MSE-optimal bandwidth h, side-specific linear fits, longitude, latitude and boundary-segment fixed effects; robust bias-corrected SE and p-value (Calonico et al., 2014). West/East: mean outcome within h of the boundary on each side. N_h: observations within h. q: sharpened FDR q-values (Anderson, 2008); rows with $q \leq 0.10$ shaded. Block-year counts (zeros filled), density per km^2 and rate per 1,000 residents ($+1$), both $\log(1+x)$ and averaged over 2017–2024; blocks within 3.5 km of the boundary. Property: theft, burglary and vehicle theft; Violent: injuries and assaults excluding homicide; Violence against women: domestic violence, sexual crimes and feminicide. q across the ten outcomes.

Back: Explanations

Cadastral land values predict housing prices



Cadastral land value per m^2 (log), Guadalajara Municipal Cadaster, 2017

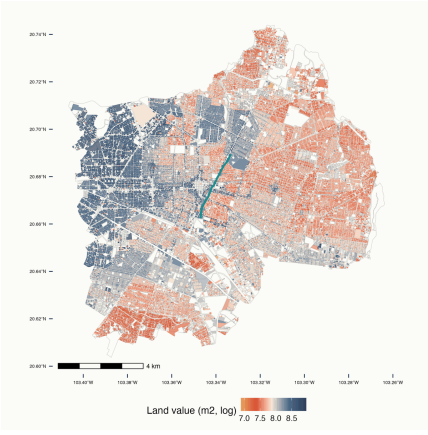


Against listed housing price per m^2, IIEG web-scraped listings, 2022; *colonia* means, log scale

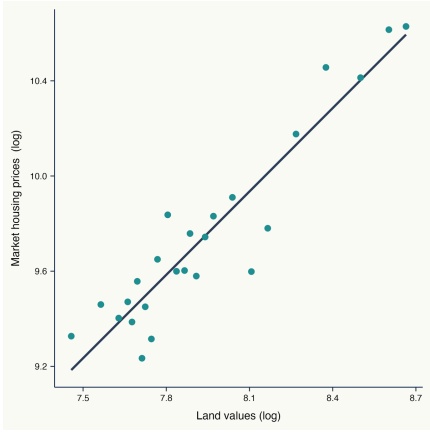
Higher land values go with proportionally higher housing prices, close to one for one: land values are a proxy for housing values

Back: Explanations By land use

Land values: the gap survives the control



Cadastral land values (log), 2017



Cadastral values track listed housing prices

Back: Explanations

Land values: the gap survives the control

Parcel-level spatial RDD on log land value per m^2, by block land use; class-specific bandwidths (Calonico et al., 2014)
longitude, latitude, boundary-segment dummies; robust bias-corrected inference, clustered by block

Block class	$\hat{\tau}$	SE	95% CI	p-val	h	N_h
Residential	-0.208	0.541	[-0.883, 1.238]	0.743	487.8	495
Mixed	-0.249	0.207	[-0.887, -0.075]	0.020	694.8	3,471
Non-residential	-0.302	0.065	[-0.487, -0.232]	0.000	811.7	7,702
Differences						
Mixed − Residential	-0.040	0.579	[-1.794, 0.477]	0.256		
Non-residential − Residential	-0.094	0.545	[-1.605, 0.531]	0.325		
Non-residential − Mixed	-0.054	0.217	[-0.304, 0.547]	0.575		

Lower land values east of the boundary for non-residential and mixed parcels; residential estimate imprecise

residential: dwellings and only neighborhood-serving establishments (DENUE, $\leq$10 workers, everyday activities)

Back: Explanations

Land values: the gap survives the control

Block-level spatial RDD, 2020 Census; years of schooling, block mean 11.22 (SD 2.00), standardized; unweighted clean controls: longitude, latitude, boundary-segment dummies; land value: log cadastral value per m^2

Specification	$\hat{\tau}$	SE	95% CI	p-val	Poly	h	b	N_h
Main (unweighted)	-0.622	0.189	[-0.992, -0.253]	0.001	1	821.3	1651.0	593
Main (Poly 2)	-0.609	0.175	[-0.952, -0.267]	0.000	2	1820.7	2859.2	1,797
With clean controls	-0.643	0.164	[-0.965, -0.321]	0.000	1	691.2	1645.9	463
With clean controls (Poly 2)	-0.733	0.167	[-1.060, -0.406]	0.000	2	1542.1	2767.6	1,397
+ Land value control	-0.402	0.143	[-0.682, -0.122]	0.005	1	804.8	1724.2	570
+ Land value control (Poly 2)	-0.483	0.145	[-0.766, -0.199]	0.001	2	1779.5	2866.4	1,687

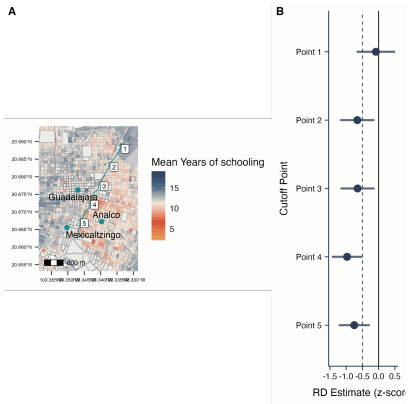
With land-value controls: 0.80–0.97 years

land values may themselves capitalize the boundary's history

Back: Explanations

Land values: the gap survives the control

Boundary-adaptive RDD with block-level land-value controls, years of schooling (standardized); 95% uniform bands (Cattaneo et al., 2025)



Pointwise estimates stay negative along the boundary

Back: Explanations

Street infrastructure is not worse east of the boundary

Block-front spatial RDD, 2020 National Housing Inventory (INV); block fronts within 3 km of the former creek

	Mean within h							
Outcome	West	East	$\hat{\tau}$	SE	p-val	q-val	h (m)	N_h
Paved street (asphalt or concrete)	0.827	0.950	0.474	0.147	0.002	0.012	468	1,713
Sidewalk	0.905	0.917	0.210	0.075	0.003	0.012	816	3,339
Wheelchair ramp	0.588	0.784	0.361	0.110	0.001	0.012	584	2,174
Storm drain	0.230	0.306	0.205	0.109	0.067	0.104	702	2,759
Street lighting	0.552	0.557	0.138	0.096	0.095	0.125	760	3,037
Trees	0.716	0.658	0.029	0.108	0.749	0.599	893	3,730
Pedestrian crossing	0.841	0.845	-0.023	0.078	0.714	0.599	421	1,525
Pedestrian traffic light	0.115	0.063	0.023	0.162	0.976	0.602	660	2,552
Public transport stop	0.071	0.087	-0.082	0.111	0.412	0.347	919	3,869
Street stall (semi-fixed)	0.063	0.062	0.123	0.146	0.239	0.212	726	2,881
Street stall (mobile)	0.061	0.078	0.279	0.129	0.011	0.024	705	2,777

Notes: Each row is a separate spatial RDD, east minus west, with MSE-optimal bandwidth h, side-specific linear fits, longitude, latitude and boundary-segment fixed effects; robust bias-corrected SE and p-value (Calonico et al., 2014). West/East: mean outcome within h of the boundary on each side. N_h: observations within h. q: sharpened FDR q-values (Anderson, 2008); rows with $q \leq 0.10$ shaded. One observation per block front; each amenity equals one if the INV records it on that front (paving: asphalt or concrete), missing when not applicable or not specified. West/East are raw shares; $\hat{\tau}$ and SE are in standard deviations of the outcome. Side and boundary segment are those of the front's census block. q across the eleven outcomes. Where fronts differ, the eastern side has more.

Back: Explanations

Urban form: same parcel size and residents per block

Block-level spatial RDD, 12,077 populated blocks; 2020 Census and 2017 cadastre (289,935 parcels assigned to blocks by centroid)

	Mean within h							
Outcome	West	East	$\hat{\tau}$	SE	p-val	q-val	h (m)	N_h
Panel A. Blocks (2020 Census, 2017 cadastre)								
Residents per block	75.3	92.1	2.273	8.911	0.978	0.432	969	735
Occupied dwellings per block	23.2	27.2	-0.740	2.966	0.601	0.430	922	682
Log median parcel area (m^2)	5.170	5.126	0.010	0.095	0.818	0.432	1,483	1,272
Share of parcels in condominium	0.009	0.007	-0.015	0.015	0.179	0.176	2,001	2,035
Panel B. Parcels (2017 cadastre; SEs clustered by block)								
Log parcel area (m^2), parcel level	5.165	5.130	0.007	0.061	0.927	–	1,456	29,102

Residents and dwellings per block, parcel size, and the share of condominiums do not differ at the boundary

Notes: Each row is a separate spatial RDD, east minus west, with MSE-optimal bandwidth h, side-specific linear fits, longitude, latitude and boundary-segment fixed effects; robust bias-corrected SE and p-value (Calonico et al., 2014). West/East: mean outcome within h of the boundary on each side. N_h: observations within h. q: sharpened FDR q-values (Anderson, 2008); rows with $q \leq 0.10$ shaded. Panel A: one observation per populated block; Panel B: one observation per parcel, SEs clustered by block. q across the ten block outcomes estimated, of which four are shown.

Back: Explanations

Business structure: smaller establishments east, same activity mix

Spatial RDD, 89,893 establishments (DENUE December 2019) placed in the nearest 2020 census block within 50 m

	Mean within h							
Outcome	West	East	$\hat{\tau}$	SE	p-val	q-val	h (m)	N_h
Panel A. Establishment characteristics (establishment level)								
0-5 employees	0.788	0.844	0.053	0.028	0.035	0.132	1,434	16,083
6-10 employees	0.114	0.087	-0.022	0.014	0.098	0.172	1,746	19,916
11-30 employees	0.066	0.050	-0.007	0.010	0.408	0.374	1,423	15,985
31+ employees	0.032	0.020	-0.024	0.010	0.006	0.072	1,232	14,077
Log employees (bracket midpoint)	1.444	1.344	-0.114	0.054	0.018	0.108	1,368	15,319
Registered legal name (formality proxy)	0.293	0.203	-0.040	0.030	0.282	0.320	2,075	23,932
Lists a phone	0.404	0.402	0.033	0.049	0.490	0.417	1,426	16,008
Lists an e-mail	0.256	0.179	0.048	0.035	0.060	0.158	897	10,032
Lists a website	0.122	0.068	-0.038	0.019	0.068	0.158	1,834	20,913
Inside market / plaza / commercial building	0.091	0.091	0.050	0.146	0.629	0.523	2,126	24,565
Semi-fixed unit	0.017	0.015	0.011	0.012	0.336	0.337	1,796	20,463
Neighborhood-serving activity	0.334	0.307	-0.000	0.044	0.805	0.675	1,593	17,809
Panel B. Density (block level)								
Block has any establishment	0.942	0.901	0.062	0.037	0.066	0.071	1,071	851
Log(1 + establishments in block)	2.089	2.050	-0.175	0.166	0.214	0.111	1,510	1,356
Establishments per 100 residents (populated blocks)	37.8	25.0	-47.301	23.581	0.018	0.059	1,745	1,677

Establishments east of the boundary are smaller ($q \approx 0.07$–0.13); the activity mix does not differ

Notes: Each row is a separate spatial RDD, east minus west, with MSE-optimal bandwidth h, side-specific linear fits, longitude, latitude and boundary-segment fixed effects; robust bias-corrected SE and p-value (Calonico et al., 2014). West/East: mean outcome within h of the boundary on each side. N_h: observations within h. q: sharpened FDR q-values (Anderson, 2008); rows with $q \leq 0.10$ shaded. Running variable: the block's signed distance; SEs clustered by block. Panel A: one observation per establishment; Panel B: one per block. q within each panel. Results hold excluding establishments inside markets and commercial buildings and adding log distance to the CBD; the household survey shows no difference in written contracts or self-employment.

Back: Explanations

Business structure: smaller establishments east, same activity mix

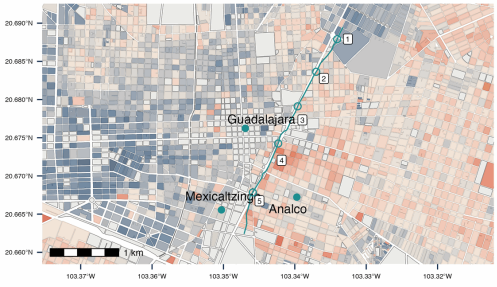
Spatial RDD, SCIAN 2-digit sector shares; same sample and specification

	Mean within h							
Outcome	West	East	$\hat{\tau}$	SE	p-val	q-val	h (m)	N_h
SCIAN 2-digit sector shares (establishment level)								
Manufacturing	0.097	0.076	-0.018	0.024	0.655	0.859	1,556	17,441
Wholesale trade	0.060	0.062	0.022	0.023	0.237	0.668	1,070	12,266
Retail trade	0.340	0.430	0.100	0.049	0.057	0.667	2,440	28,629
Information	0.003	0.001	0.001	0.002	0.545	0.859	1,632	18,265
Finance and insurance	0.017	0.010	-0.013	0.012	0.129	0.667	2,276	26,483
Real estate and rental	0.012	0.008	-0.003	0.005	0.732	0.859	2,305	26,811
Professional services	0.041	0.010	-0.009	0.010	0.644	0.859	1,245	14,141
Business support services	0.019	0.011	0.010	0.006	0.048	0.667	1,687	19,077
Education	0.025	0.016	-0.007	0.006	0.143	0.667	1,500	16,776
Health and social assistance	0.037	0.026	-0.002	0.007	0.719	0.859	1,087	12,469
Recreation	0.010	0.009	-0.001	0.003	0.665	0.859	2,554	30,482
Food and lodging	0.138	0.104	-0.006	0.028	0.967	0.936	1,741	19,801
Other services (repair, personal)	0.131	0.151	-0.028	0.026	0.198	0.667	1,382	15,493
Government	0.015	0.004	-0.013	0.009	0.092	0.667	2,244	26,040

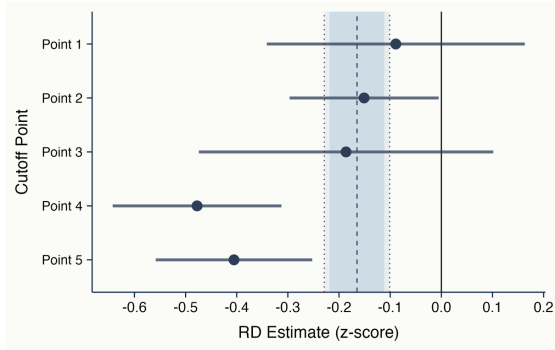
Notes: Each row is a separate spatial RDD, east minus west, with MSE-optimal bandwidth h, side-specific linear fits, longitude, latitude and boundary-segment fixed effects; robust bias-corrected SE and p-value (Calonico et al., 2014). West/East: mean outcome within h of the boundary on each side. N_h: observations within h. q: sharpened FDR q-values (Anderson, 2008); rows with $q \leq 0.10$ shaded. One observation per establishment; SEs clustered by block. q across the 14 sectors; none is shaded (smallest $q = 0.67$; retail is the only sector with $p < 0.10$).

Back: Explanations

The gap is largest where Analco faced the colonial settlements



199,553 residents in the area shown; blank blocks, suppressed to protect confidentiality, hold 1,033 of them

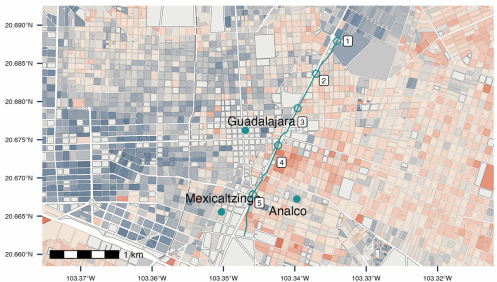


Years of schooling: **points 4–5:** -2.3 **and** -1.9 **years**

Point 5: Analco vs. Mexicaltzingo; point 4: Analco vs. Spanish Guadalajara; points 1–3: farther north

Table | Other outcomes along the boundary | ◂ Back

The gap is largest where Analco faced the colonial settlements



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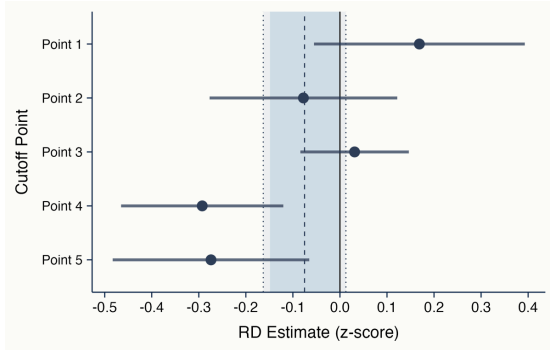
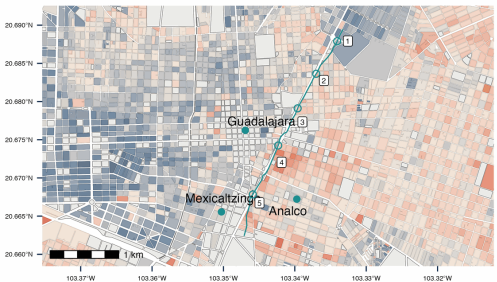


Attendance: **points 4–5:** −12.6 **and** −11.8 **pp**

Point 5: Analco vs. Mexicaltzingo; point 4: Analco vs. Spanish Guadalajara; points 1–3: farther north

Table | Other outcomes along the boundary | ◂ Back

The gap is largest where Analco faced the colonial settlements



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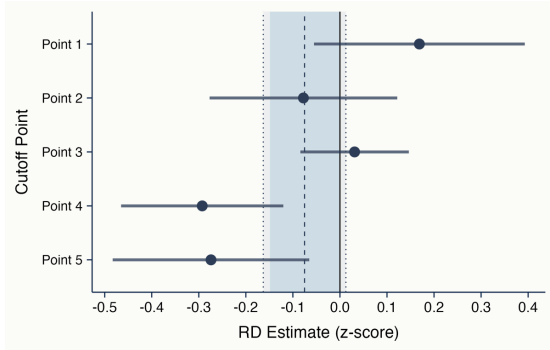


Attendance: **points 4–5:** −12.6 **and** −11.8 **pp**

Point 5: Analco vs. Mexicaltzingo; point 4: Analco vs. Spanish Guadalajara; points 1–3: farther north

Same gap facing Spaniards or Indigenous allies: not ancestry, not the avenue

Table | Other outcomes along the boundary | ◂ Back

The gap is larger at the two southern points than at the three northern ones

2020 Census, individual level; robust bias-corrected estimates, 95% uniform bands; q-values across the six contrasts within each outcome

	Years of schooling		School attendance	
	Estimate (s.e.)	95% uniform band	Estimate (s.e.)	95% uniform band
Panel A. Pointwise estimates along the boundary				
Point 1 — North	−0.427 (0.457)	[−1.60, 0.75]	0.073 (0.038)	[−0.02, 0.17]
Point 2 — North	−0.724 (0.264)	[−1.40, −0.04]	−0.033 (0.034)	[−0.12, 0.05]
Point 3 — North	−0.894 (0.522)	[−2.23, 0.45]	0.013 (0.020)	[−0.04, 0.06]
Point 4 — South (Mexicaltzingo)	−2.291 (0.299)	[−3.06, −1.52]	−0.126 (0.030)	[−0.20, −0.05]
Point 5 — South (Mexicaltzingo)	−1.945 (0.277)	[−2.66, −1.23]	−0.118 (0.036)	[−0.21, −0.03]
Panel B. Pairwise differences, southern segment minus each northern point				
	Difference (s.e.)	q-value	Difference (s.e.)	q-value
Point 4 − point 1	−1.863*** (0.547)	0.002	−0.198*** (0.048)	0.000
Point 4 − point 2	−1.567*** (0.399)	0.001	−0.092** (0.045)	0.016
Point 4 − point 3	−1.397** (0.602)	0.008	−0.139*** (0.035)	0.000
Point 5 − point 1	−1.517*** (0.535)	0.004	−0.190*** (0.052)	0.000
Point 5 − point 2	−1.221*** (0.383)	0.002	−0.084* (0.049)	0.029
Point 5 − point 3	−1.051* (0.591)	0.025	−0.131*** (0.041)	0.001

Standard errors of differences treat points as independent: `rd2d` does not export their covariance

Back: Along the boundary

Along the boundary, crowding and ethnic composition share the southern gradient; land values, assets, crime, schools and businesses do not

Block-level boundary RDD at the same five points, east minus west, each dimension standardized; south (points 4–5) minus north (points 1–3)

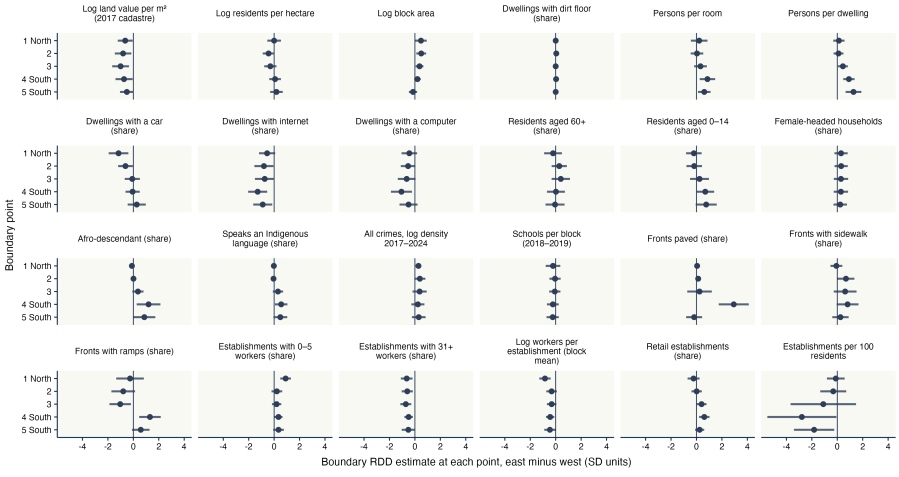
	Boundary estimate (SD)		South − North				
Block-level dimension	North (1–3)	South (4–5)	Diff.	SE	*p*-val	*q*-val	Pattern
Land and urban form							
Log land value per m^2 (2017 cadastre)	-0.82	-0.62	0.19	0.29	0.498	0.597	no gradient
Log residents per hectare	-0.25	0.13	0.38	0.22	0.095	0.128	differs, no clear valence
Log block area	0.45	0.03	-0.42	0.16	0.007	0.030	differs, no clear valence
Housing and households (2020 Census)							
Dwellings with dirt floor (share)	0.02	0.02	-0.00	0.02	0.994	0.845	no gradient
Persons per room	0.19	0.74	0.55	0.26	0.033	0.072	east worse in the south
Persons per dwelling	0.22	1.10	0.87	0.23	0.000	0.002	differs, no clear valence
Dwellings with a car (share)	-0.42	0.11	0.73	0.30	0.016	0.053	east better in the south
Dwellings with internet (share)	-0.70	-1.10	-0.40	0.34	0.248	0.297	no gradient
Dwellings with a computer (share)	-0.54	-0.78	-0.24	0.33	0.475	0.597	no gradient
Demographics (2020 Census)							
Residents aged 60+ (share)	0.16	-0.02	-0.17	0.33	0.597	0.608	no gradient
Residents aged 0–14 (share)	-0.05	0.73	0.78	0.34	0.022	0.063	differs, no clear valence
Female-headed households (share)	0.30	0.26	-0.04	0.25	0.879	0.845	no gradient
Afro-descendant (share)	0.09	1.03	0.94	0.34	0.005	0.027	differs, no clear valence
Speaks an Indigenous language (share)	0.08	0.52	0.44	0.20	0.030	0.072	differs, no clear valence
Crime							
All crimes, log density 2017–2024	0.35	0.27	-0.08	0.23	0.737	0.729	no gradient
School supply							
Schools per block (2018–2019)	-0.11	-0.23	-0.12	0.22	0.597	0.608	no gradient
Street infrastructure (INV 2020)							
Fronts paved (share)	0.14	1.37	1.23	0.38	0.001	0.009	east better in the south
Fronts with sidewalk (share)	0.41	0.53	0.12	0.34	0.725	0.729	no gradient
Fronts with ramps (share)	-0.70	0.95	1.64	0.40	0.000	0.001	east better in the south
Businesses (DENUE 2019)							
Establishments with 0–5 workers (share)	0.44	0.34	-0.09	0.18	0.598	0.608	no gradient
Establishments with 31+ workers (share)	-0.65	-0.50	0.15	0.21	0.467	0.597	no gradient
Log workers per establishment (block mean)	-0.49	-0.45	0.05	0.19	0.789	0.756	no gradient
Retail establishments (share)	0.06	0.43	0.37	0.19	0.047	0.087	differs, no clear valence
Establishments per 100 residents	-0.51	-2.30	-1.79	0.94	0.056	0.095	differs, no clear valence

Notes: Location-based boundary RDD (Cattaneo et al., 2025) on populated census blocks at the five boundary points of the main estimates; dimensions standardized (SD). North, South: means of the pointwise estimates; Diff.: South − North, SE under independence across points; *q*: sharpened FDR across 24 dimensions (Anderson, 2008), $q \leq 0.10$ shaded. Pattern "east worse in the south": the sign would leave the eastern side worse off at points 4–5 than at 1–3, the direction that could confound the historical contrast. Ethnic and age composition may themselves be colonial legacies. The pooled individual-level gap survives controls for Afro-descendant and Indigenous-language status, dwelling class, migration and activity (-0.117 vs. -0.141 SD) and, for attendance, parental schooling and household size (-0.086 SD); the point-by-point estimates control for age and sex only.

Back: Along the boundary

Along the boundary, crowding and ethnic composition share the southern gradient; land values, assets, crime, schools and businesses do not

Pointwise estimates, robust bias-corrected 95% CIs



Back: Along the boundary

Income: the gap runs through schooling

2015 Intercensal Survey, adults 18+: side-specific linear trends within 5 km, no optimal bandwidth; SEs clustered by household

	Monthly income (log)		
	(1)	(2)	(3)
$\mathbb{1}\{East_i\}$	-0.084***	-0.010	0.075
	(0.027)	(0.024)	(0.068)
Years of schooling (YoS)		0.065***	0.068***
		(0.001)	(0.005)
$\mathbb{1}\{East_i\} \times$ YoS			-0.006
			(0.006)
Sample (age in years)	18+	18+	18+
Individual controls	✓	✓	✓
Geographic controls	✓	✓	✓
Observations	31,584	31,478	31,478
Dep. var. mean	8.65	8.65	8.65
Dep. var. SD	0.644	0.645	0.645
R^2	0.123	0.268	0.269

Income gap $-8.1\% \rightarrow -1\%$ once schooling enters; returns to schooling similar across sides

Back: Expected returns

Trust in institutions is lower on the east side

Life Quality Survey 2022, survey weights; sharpened q-values across social-capital outcomes within each panel

	Institutional trust index (1)	Trust in churches (2)	Trust in local congress (3)	Trust in electoral institute (4)	Trust in political parties (5)	Trust in government (6)	Trust in judiciary (7)	Trust in police (8)	Trust in army (9)	Trust in navy (10)	Trust in prosecutor's office (11)
Panel A. Without schooling control											
$\mathbb{1}\{East\}$	-0.235***	-0.111	-0.182**	-0.234**	-0.178***	-0.259***	-0.195**	-0.259***	-0.380***	-0.480***	-0.111
	(0.064)	(0.116)	(0.081)	(0.103)	(0.063)	(0.075)	(0.077)	(0.088)	(0.111)	(0.112)	(0.085)
q-value	[0.004]	[0.328]	[0.049]	[0.049]	[0.018]	[0.005]	[0.035]	[0.017]	[0.005]	[0.001]	[0.221]
R^2	0.093	0.105	0.048	0.061	0.059	0.062	0.070	0.050	0.122	0.130	0.055
Panel B. Controlling for years of schooling											
$\mathbb{1}\{East\}$	-0.232***	-0.147	-0.182**	-0.225**	-0.178***	-0.258***	-0.187**	-0.252***	-0.370***	-0.466***	-0.105
	(0.065)	(0.114)	(0.083)	(0.105)	(0.062)	(0.075)	(0.077)	(0.089)	(0.112)	(0.113)	(0.087)
Years of schooling	0.002	-0.019***	0.000	0.005	0.000	0.000	0.004	0.004	0.005	0.008	0.003
	(0.003)	(0.006)	(0.005)	(0.005)	(0.003)	(0.004)	(0.004)	(0.005)	(0.005)	(0.005)	(0.004)
q-value	[0.006]	[0.230]	[0.056]	[0.058]	[0.019]	[0.006]	[0.041]	[0.019]	[0.007]	[0.002]	[0.249]
R^2	0.094	0.139	0.048	0.064	0.059	0.062	0.073	0.053	0.125	0.136	0.057
Observations	289	289	289	289	289	289	289	289	289	289	289
Dep. var. mean	0.229	0.472	0.164	0.265	0.142	0.178	0.199	0.203	0.349	0.373	0.204
Age-Sex fixed effects	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Survey weights	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓

Back: City survey

Memberships in organizations do not differ across sides

Life Quality Survey 2022, survey weights; sharpened q-values across social-capital outcomes within each panel

	Member of any organization (1)	Number of memberships (2)	Member of neighborhood assoc. (3)	Member of school-related org. (4)	Member of political org. (5)	Member of religious group (6)	Member of NGO/ assistance org. (7)
Panel A. Without schooling control							
$\mathbb{1}\{East\}$	-0.086	-0.088	0.066	-0.039	0.023	-0.076	-0.062
	(0.100)	(0.188)	(0.056)	(0.050)	(0.041)	(0.085)	(0.063)
q-value	[0.329]	[0.443]	[0.268]	[0.347]	[0.399]	[0.329]	[0.328]
R^2	0.068	0.046	0.033	0.034	0.029	0.068	0.050
Panel B. Controlling for years of schooling							
$\mathbb{1}\{East\}$	-0.058	-0.041	0.080	-0.032	0.032	-0.064	-0.057
	(0.101)	(0.188)	(0.057)	(0.050)	(0.044)	(0.087)	(0.063)
Years of schooling	0.015***	0.025***	0.008**	0.004**	0.004	0.007	0.002
	(0.005)	(0.009)	(0.003)	(0.002)	(0.004)	(0.004)	(0.002)
q-value	[0.404]	[0.480]	[0.200]	[0.404]	[0.404]	[0.404]	[0.361]
R^2	0.098	0.073	0.054	0.046	0.037	0.079	0.054
Observations	289	289	289	289	289	289	289
Dep. var. mean	0.114	0.178	0.041	0.018	0.036	0.065	0.018
Age-Sex fixed effects	✓	✓	✓	✓	✓	✓	✓
Survey weights	✓	✓	✓	✓	✓	✓	✓

Back: City survey

Neighbor trust and schooling: heterogeneity and falsification

Life Quality Survey 2022, survey weights; outcome in columns (4)–(9): years of schooling

	Years of schooling	Trust in neighbors	Neighbor vs. nonlocals trust	Years of schooling					
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
$1\{East\}$	-1.86	0.368**	0.093	-0.798	0.172	-1.28	-2.32**	-0.761	-1.81
	(1.13)	(0.151)	(0.064)	(1.23)	(1.73)	(1.25)	(1.17)	(1.07)	(1.12)
Years of schooling		-0.002	-0.004						
		(0.007)	(0.003)						
Trust in neighbors				0.796					
				(0.691)					
Trust in neighbors × East				-1.74*					
				(0.945)					
Neighbor vs. nonlocals trust					0.703				
					(1.65)				
Neighbor vs. nonlocals trust × East					-3.14				
					(2.19)				
Institutional trust index						1.67			
						(1.68)			
Institutional trust index × East						-1.89			
						(2.15)			
Tract-agg. neighbor trust (z)							0.775**		
							(0.310)		
Tract-agg. neighbor trust (z) × East							-1.36***		
							(0.464)		
Tract-agg. schooling (z)								1.50***	
								(0.268)	
Tract-agg. schooling (z) × East								0.042	
								(0.572)	
Tract-agg. land value (log, z)									1.22***
									(0.258)
Tract-agg. land value (log, z) × East									-0.720
									(0.719)
Observations	289	289	289	289	289	289	289	289	289
Dep. var. mean	11.33	0.6549	0.5953	11.33	11.33	11.33	11.33	11.33	11.33
R^2	0.2246	0.0887	0.0812	0.2336	0.2336	0.2277	0.2476	0.3245	0.2767
Age-Sex fixed effects	✓	✓	✓	✓	✓	✓	✓	✓	✓

Back: City survey

Static model: invest if the return beats the exchange put at risk

Elements

- Value of local exchange over outside channels: $\Delta(q, Z) = R - Z(1 - q)$
  q: share of neighbors who participate; Z: outside channels, worse with exclusion (Kranton, 1996)
- Participation $\theta_i \in \{0, 1\}$ costs $c(y_i)$: be available, give back (González de la Rocha, 1999)
- Educating a child of age n forfeits a share π_n of the network's value, rising with n (Munshi and Rosenzweig, 2006)
- β_i: use of the network; $a(s)$: weight on it, rising with salience (Gabaix, 2019)

◂ Back

Decision

$$U_i = (\rho - C_i)\, e_i + \theta_i \left[(1 - \pi_n e_i)\, \beta_i\, \Delta - c(y_i) \right]$$

$$\text{invest iff} \quad \rho - C_i \;\geq\; a(s)\, \Pi_i, \qquad \Pi_i = \theta_i\, \pi_n\, \beta_i\, \Delta$$

Predictions

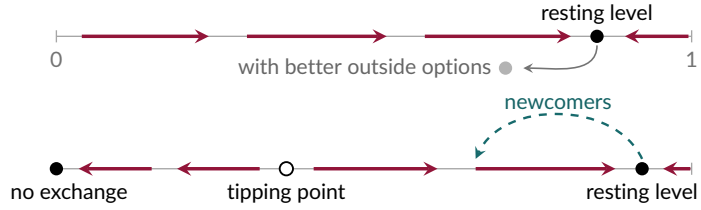
1. Salience lowers demand only where $\Pi_i > 0$
2. More use of the network (β_i), lower demand
3. The poorest cannot afford to participate ($\theta_i = 0$): no response
4. Same return ρ, higher required return
5. Larger gap in adolescence (π_n rises with n)

Dynamic model: parents pass on participation, not values

- Parents who take part in the exchange with neighbors try to pass that on, and so do parents who stay out; a child who is taught neither copies a neighbor at random (Bisin and Verdier, 2001)
- Parents try harder when what they pass on is worth more and when households like theirs are few

(a) Valuable among few participants
one resting level, which falls gradually as outside options improve

(b) Valuable only with enough participants
above the tipping point, participation sustains itself, even after outside options catch up



Lines: share of neighbors who take part, from 0 to 1. Arrows: where it moves from one generation to the next

Turnover: newcomers pull the share toward their own; it drifts back unless it falls below the tipping point

Law of motion: $q_{t+1} - q_t = k\, q_t(1-q_t)\big[(1-q_t)\lambda\Delta(q_t, Z) - q_t\, b(Z)\big]$, with $\Delta(q, Z) = R - Z(1-q)$; (a) $R > Z$, (b) $R < Z$. Turnover: $q_{t+1} = (1-\mu)\tilde{q}_{t+1} + \mu\, q^M$. Human capital is transmitted in parallel (Cunha and Heckman, 2007).

◂ Back to framework ◂ Back to turnover ◂ Back to interpretation

Parents chose between an hour of tutoring and a gift card, in eleven rows

Multiple price list (Jack et al., 2022)

Row	Amazon gift card	Math tutoring
1	MXN 0	1 hour
2	MXN 25	1 hour
3	MXN 50	1 hour
⋮	⋮	⋮
10	MXN 225	1 hour
11	MXN 250	1 hour

row order and left/right side randomized

- Practice round: a pen vs. marzipan candies
- Four comprehension checks; mean 3.8 of 4
- Binding row drawn from a sealed envelope
- 1 in 10 households paid; MXN 100 to everyone
- Outcomes: share of tutoring choices; latent WTP, two-step

Back: Design

Map-First lowers the tutoring share and willingness to pay on the east side

	Share of tutoring choices (OLS)			Willingness to pay (MXN pesos) (Jack et al. (2022), two-step latent WTP)		
	(1)	(2)	(3)	(4)	(5)	(6)
$\mathbb{1}\{Map\text{-}First_i\}$	-0.077^{*}	0.144	-0.117^{**}	-44.7	97.6	-70.9^{**}
Robust	(0.049)	(0.126)	(0.053)	(36.8)	(93.4)	(39.6)
Randomization inference	(0.046)	(0.117)	(0.051)	(30.7)	(80.8)	(33.4)
$\mathbb{1}\{East_i\}$		0.006			-18.4	
Robust		(0.219)			(172.7)	
Randomization inference		—			—	
$\mathbb{1}\{Map\text{-}First_i\} \times \mathbb{1}\{East_i\}$		-0.267^{**}			-172.2^{*}	
Robust		(0.137)			(102.9)	
Randomization inference		(0.129)			(88.4)	
Sample	Full	Full	East side	Full	Full	East side
Enumerator FE	✓	✓	✓	✓	✓	✓
Day FE	✓	✓	✓	✓	✓	✓
LASSO controls	✓	✓	✓	✓	✓	✓
Distance to boundary	✓	✓	✓	✓	✓	✓
p-value: east-side effect = west-side effect (robust / RI)		0.052 / 0.037			0.094 / 0.051	
Dep. var. mean	0.548	0.548	0.538	164.0	164.0	155.6
Control mean (Map-After)	0.561	0.561	0.573	171.6	171.6	176.3
Control SD (Map-After)	0.384	0.384	0.372	255.9	255.9	249.5
Participants	239	239	193	239	239	193
Within R^2	0.235	0.251	0.216	0.222	0.238	0.208

Stars use randomization-inference p-values; WTP standard errors from a subject-level Bayesian bootstrap of both steps

Back: Results

No covariate differs across arms

239 households; randomization-inference p-values within side; sharpened FDR q-values across all tests (Anderson, 2008)

	Map-After (control)	Map-First (treatment)	Difference (T − C)	p-value	q-value	N
Panel A. Measured before the randomized block						
$\mathbb{1}\{East_i\}$	0.840	0.782	-0.058	—	—	239
Distance to boundary (m)	1012	959	-53	0.902	1.000	239
Parent age	44.93	43.53	-1.40	0.327	1.000	239
Female respondent	0.764	0.699	-0.065	0.302	1.000	239
Parent completed high school	0.594	0.586	-0.008	1.000	1.000	239
Father completed high school	0.344	0.319	-0.026	0.766	1.000	203
Mother completed high school	0.265	0.283	0.019	0.767	1.000	229
Has partner	0.660	0.684	0.024	0.783	1.000	239
Adults in household	2.87	2.82	-0.05	0.800	1.000	239
Minors in household	1.78	1.80	0.01	0.957	1.000	239
Rooms in dwelling	2.73	2.72	-0.00	1.000	1.000	239
Crowding (persons/room)	1.79	1.89	0.10	0.311	1.000	239
Children aged 6–20	1.66	1.65	-0.01	0.937	1.000	239
Respondent works	0.651	0.692	0.041	0.574	1.000	239
Main income is a formal salary	0.755	0.729	-0.025	0.771	1.000	239
Price-list comprehension (0–4)	3.81	3.86	0.05	0.423	1.000	239

No variable differs across arms individually or after FDR correction

Back: Design Back: Results

No covariate differs across arms

239 households; randomization-inference p-values within side; sharpened FDR q-values across all tests (Anderson, 2008)

	Map-After (control)	Map-First (treatment)	Difference (T − C)	p-value	q-value	N
Panel A (cont.). Household income bracket						
Income: under $6,000	0.149	0.117	-0.032	0.541	1.000	214
Income: $6,001–9,000	0.149	0.125	-0.024	0.689	1.000	214
Income: $9,001–12,000	0.160	0.192	0.032	0.597	1.000	214
Income: $12,001–15,000	0.128	0.167	0.039	0.446	1.000	214
Income: $15,001–18,000	0.085	0.083	-0.002	1.000	1.000	214
Income: $18,001–22,000	0.074	0.158	0.084	0.088	1.000	214
Income: $22,001–27,500	0.106	0.050	-0.056	0.185	1.000	214
Income: $27,501–35,000	0.085	0.075	-0.010	0.798	1.000	214
Income: $35,001–55,000	0.053	0.017	-0.037	0.253	1.000	214
Income: over $55,000	0.011	0.017	0.006	1.000	1.000	214
Panel B. Measured after the randomized block						
Focal child female	0.425	0.414	-0.011	0.895	1.000	239
Focal child age	12.80	12.40	-0.40	0.447	1.000	239
Focal child attends school	0.896	0.925	0.029	0.502	1.000	239
Chose child: needs academic help	0.094	0.090	-0.004	1.000	1.000	239
Chose child: ability/aspiration	0.094	0.053	-0.042	0.304	1.000	239
Chose child: age/only eligible	0.321	0.429	0.108	0.105	1.000	239

No variable differs across arms individually or after FDR correction

Back: Design Back: Results

The east-side effect is not confined to poorer or less-educated households

East side ($N = 193$): share of tutoring choices

	Map-First effect below	Map-First effect above	Difference RI p
Income bracket above median	-0.072 (0.074)	-0.176^{*} (0.090)	0.35
A parent completed high school	-0.093 (0.096)	-0.129^{**} (0.059)	0.76
Both	-0.059 (0.071)	-0.207^{**} (0.093)	0.21
Local ties index above median	-0.128^{*} (0.075)	-0.113 (0.071)	0.88

Share of tutoring choices; controls as in the main table; HC1 SEs; RI p for the difference, 2,000 permutations; MDE ≈ 0.3

Income and parental schooling barely differ across the boundary, so they cannot account for the concentration on the east side

Local-ties index built from survey modules asked after the choice

Back: Results Back: Survey modules

The east-side effect is not confined to poorer or less-educated households

East side ($N = 193$): willingness to pay (MXN)

	Map-First effect below	Map-First effect above	Difference RI p
Income bracket above median	−49.3 (48.5)	−98.2 (60.4)	0.53
A parent completed high school	−65.1 (61.1)	−74.5* (39.6)	0.90
Both	−41.3 (46.3)	−117.4* (63.1)	0.33
Local ties index above median	−76.6 (52.9)	−71.0 (44.1)	0.94

Latent WTP (Jack et al., 2022); HC1 SEs, not corrected for the first step; RI p for the difference, 2,000 permutations; MDE ≈ MXN 200

Income and parental schooling barely differ across the boundary, so they cannot account for the concentration on the east side

Local-ties index built from survey modules asked after the choice

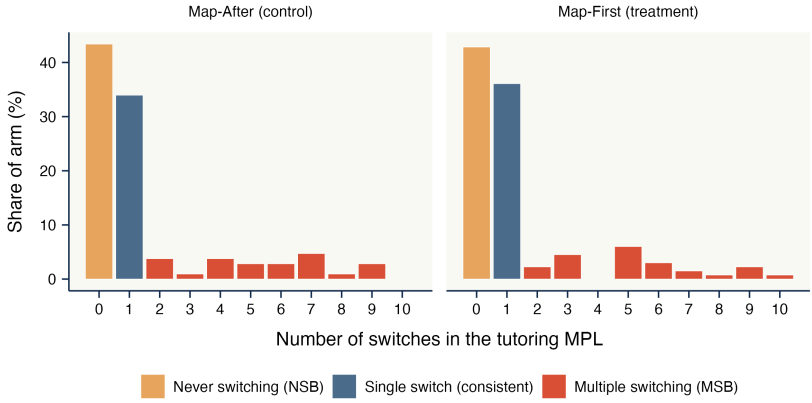
Back: Results Back: Survey modules

Comprehension and use of the choice list do not differ across arms

	Map-After (control)	Map-First (treatment)	All	Difference p-value
Comprehension score (0–4)	3.81	3.86	3.84	0.423
Passed all comprehension checks	0.840	0.902	0.874	0.184
Consistent on the practice MPL	0.547	0.624	0.590	0.250
Never-switching (NSB)	0.434	0.429	0.431	1.000
Multiple-switching (MSB)	0.226	0.211	0.218	0.874
Takes the $0 card at a zero price	0.179	0.203	0.192	0.752
Enumerators / field days	6 enumerators, 15 days; treatment $\perp$ enumerator: $F = 1.45$, $p = 0.215$			

Back: Design Back: Results

Comprehension and use of the choice list do not differ across arms



Back: Design Back: Results

No single participant drives the estimates

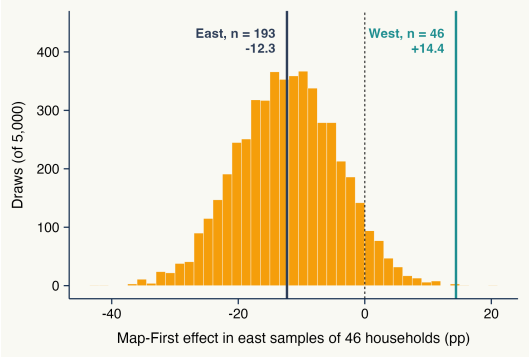
239 participants; min and max over the 239 re-estimations dropping one participant
enumerator and day FE, LASSO controls, east-by-distance terms

	Estimate	Leave-one-out range
Panel A. Share of tutoring choices		
Pooled effect (H1)	-0.077	$[-0.090,\ -0.067]$
Map-First $\times$ East (H2)	-0.267	$[-0.297,\ -0.195]$
Priming effect on the east side	-0.117	$[-0.130,\ -0.103]$
Panel B. Willingness to pay (MXN pesos)		
Pooled effect (H1)	-44.7	$[-53.1,\ -38.5]$
Map-First $\times$ East (H2)	-172.2	$[-195.3,\ -127.8]$
Priming effect on the east side	-70.9	$[-78.0,\ -62.3]$

Back: Results

East samples as small as the west's almost always show a negative effect

Draw 46 east households at random, as many as in the west and with the same split between arms, estimate the Map-First effect, and repeat 5,000 times



Even with a sample this small, the two sides give different signals: 94% of the draws are negative and 4 of 5,000 reach the west's +14 pp

Share of tutoring choices. Controls and fixed effects of the main specification, estimated on the full sample. Re-estimated within each sample of 46: 78% of the draws negative, 4% at or above +14 pp. Willingness to pay: 92% negative.

Back: Results

A bounded measure of willingness to pay gives the same sign

239 participants; controls as in the main table
two-sided *p*-values in parentheses, subject-level Bayesian bootstrap repeating both steps (1,000 replications)

	Pooled (H1)	Map-First × East (H2)	Effect on the east side
Latent scaled-probit WTP (two-step, main text)	−44.7 (0.146)	−172.2 (0.051)	−74.6 (0.025)
Area under demand ($0–$250, participant RE)	−18.4 (0.130)	−67.7 (0.047)	−30.2 (0.022)

The MXN 250 ceiling compresses the bounded measure; same sign, east-side effect significant

Back: Results

Guessing the purpose of the study does not explain the east-side effect

"What do you think the study is about?" — 30 of 239 infer a neighborhood–education link

interaction specification of the main table; HC1 SEs; significance by randomization inference within side

Panel A. Robustness of the priming effect on the east side

	(1)	(2)	(3)
Share of tutoring choices	−0.117**	−0.124**	−0.124**
[Map-First + Map-First×East]	(0.052)	(0.051)	(0.056)
Willingness to pay, latent (pesos)	−71.1**	−75.3**	−78.8**
[Map-First + Map-First×East]	(33.9)	(33.3)	(36.6)
Willingness to pay, bounded area (pesos)	−28.6**	−30.6**	−30.6**
[Map-First + Map-First×East]	(13.6)	(13.3)	(14.7)
Guess-theme controls		✓	
Sample	All	All	Non-guessers
Participants	239	239	209

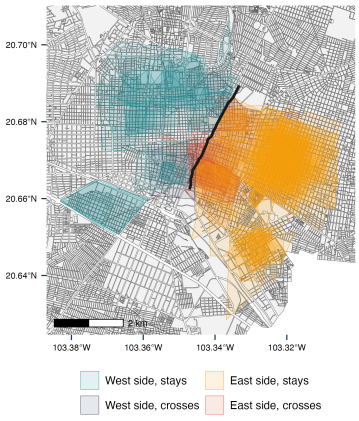
Panel B. Effect of the treatment on the inferred topic (linear probability models)

	Map-First	Map-First × East	Dep. var. mean
Guessed mechanism	0.217	−0.224*	0.126
	(0.102)	(0.111)	
Mentioned neighborhood	0.178	−0.131	0.360
	(0.139)	(0.155)	
Participants		239	

The inference rises on the west side, where no effect appears; east-side estimate unchanged

Back: Results

What the map makes salient: a compact area around home



Geometry of the drawn neighborhood

	All	East side	West side
Barrio area (km^2), median	0.58	0.56	0.71
Barrio area (km^2), mean	0.79	0.77	0.91
Home inside the drawn barrio	0.876	0.862	0.933
Home off-centering (dist./$\sqrt{\text{area}}$)	0.40	0.39	0.43
Crosses the historical boundary	0.112	0.122	0.067
Vertices drawn	4.3	4.3	4.4
Participants	233	188	45

What guided the drawing (Haaland et al., 2025)

	All	East side	West side
Physical extent	0.62	0.63	0.61
Everyday use	0.49	0.50	0.48
Daily routine or familiarity	0.37	0.37	0.39
Errands, shops, work, or school	0.19	0.20	0.13
People or belonging	0.13	0.15	0.07
Social ties	0.10	0.11	0.04
Long residence or family roots	0.04	0.04	0.02
Safety or calm	0.06	0.05	0.11
Participants	239	193	46

Back: Results

The boundary prime changes explanations on the east side only

Adjusted: Map-First assignment, interview duration, sex, age, age^2
randomization inference, 50,000 permutations within side; sharpened q-values within panel

	Share mentioning		Prime effect		Adjusted	
	Not primed	Boundary primed	Effect	q	Effect	q
Panel A. East side, historically excluded ($N = 193$)						
Economic	0.905	0.755	-0.150^{***}	0.020	-0.170^{***}	0.005
Neighborhood	0.084	0.143	0.059	0.350	0.067	0.170
Family	0.095	0.143	0.048	0.395	0.039	0.320
Individual	0.316	0.357	0.041	0.637	0.044	0.320
Neighborhood − economic			0.209^{***}	0.020	0.237^{***}	0.005
Panel B. West side ($N = 46$)						
Economic	0.880	0.857	−0.023	1.000	−0.107	0.791
Neighborhood	0.200	0.143	−0.057	1.000	−0.129	0.791
Family	0.160	0.238	0.078	1.000	0.126	0.791
Individual	0.480	0.286	−0.194	1.000	−0.149	0.791
Neighborhood − economic			−0.034	1.000	−0.022	0.791

Back: Explanations ◂ Back

Within the sample, income, family schooling, and roots in the colonia do not differ across sides

	Mean west side	Mean east	East coefficient	q
Panel G. Household income, family schooling, and roots in the colonia				
Monthly household income†	17,400	15,705	−0.100 (0.140)	1.00
Monthly household income per member†	4,149	3,824	−0.065 (0.148)	1.00
Household size	4.48	4.67	0.10 (0.24)	1.00
Years of schooling, more-schooled parent	13.15	12.40	−0.69 (0.52)	1.00
A parent completed high school	0.739	0.679	−0.050 (0.071)	1.00
Years of schooling, respondent's father	9.58	8.08	−1.83** (0.90)	0.89
Years of schooling, respondent's mother	7.77	8.08	0.16 (0.71)	1.00
Schooling gain over the more-schooled grandparent (years)	3.52	3.27	0.04 (0.77)	1.00
Share of close relatives living in the colonia	0.141	0.196	0.050 (0.036)	1.00
Years lived in the colonia	25.70	25.17	−0.74 (2.74)	1.00
A parent lived in the colonia	0.478	0.580	0.091 (0.083)	1.00

Controls: Map-First, sex, age, age^2; no respondent-education FE. Income at bracket midpoints; relatives share from five categories (0–1). HC1 SEs. † Poisson PML, semi-elasticities. Sharpened q-values within panel.

Back: Survey modules ◂ Back

Explicit norms and attachment do not differ across sides

	Mean west side	Mean east	East coefficient	q
Panel E. Perceived neighbor norms (pluralistic ignorance)				
Personal norm: youth should work, not study (1–7)	2.61	2.55	−0.237 (0.311)	1.00
Perceived neighbors sharing that norm (0–10)	5.57	5.70	0.163 (0.355)	1.00
Pluralistic ignorance (perceived − own, both on 0–10)	2.88	3.11	0.558 (0.636)	1.00
Panel F. Stated attachment to the neighborhood				
Feels part of this colonia (1–7)	5.76	5.72	−0.137 (0.253)	1.00
Would prefer to live in another colonia (1–7)	3.22	3.16	−0.077 (0.437)	1.00
Makes an effort so children know the neighbors (1–7)	4.48	4.92	0.258 (0.377)	1.00

Survey sample, $N = 238$

Back

Magnitude checks: different estimates imply similar magnitudes

Back: Results ◂ Back

Magnitude checks: different estimates imply similar magnitudes

Check 1

How large is the educational tradeoff associated with the neighborhood?

Survey threshold compared with the experimental change in willingness to pay

Check 2

What does the survey threshold imply for years of schooling?

Share of east-side parents at the margin compared with the census schooling gap

Back: Results ◂ Back

Magnitude checks: different estimates imply similar magnitudes

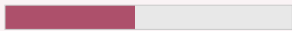
How large is the educational tradeoff associated with the neighborhood?

Survey

East-side parents require an extra MXN 5,300 before university is worthwhile

45%

of the expected university premium



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Back: Results ◂ Back

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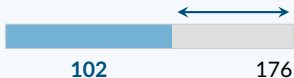
Experiment

Map-First lowers tutoring WTP from MXN 176 to MXN 102

−42%

change in willingness to pay



Back: Results ◂ Back

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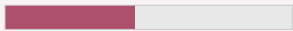

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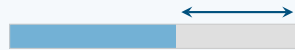

45% ≈ **42%**

Different exercises, same order of magnitude: about two fifths

Magnitude comparison, not a structural estimate: the survey and experiment use different goods, horizons, and contrasts.

Back: Results ◂ Back

Magnitude checks: different estimates imply similar magnitudes

What does the survey threshold imply for years of schooling?

Among east-side parents, compare expected university earnings with the east-side and west-side thresholds

18% expect university earnings to clear their own threshold

Back: Results Back

Magnitude checks: different estimates imply similar magnitudes

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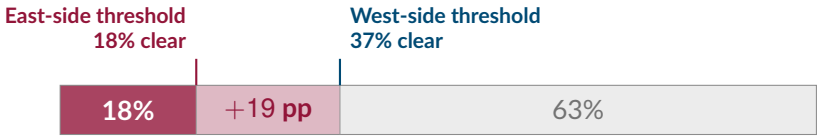
Among east-side parents, compare expected university earnings with the east-side and west-side thresholds



The additional **19 pp** clear the west-side threshold but not their own

Back: Results ◂ Back

Magnitude checks: different estimates imply similar magnitudes

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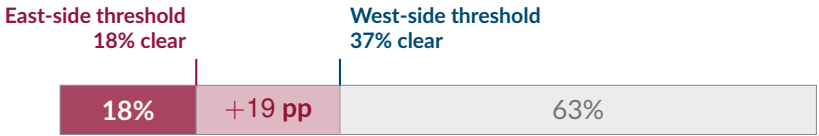
Among east-side parents, compare expected university earnings with the east-side and west-side thresholds



If this marginal 19% obtained four additional years of university:

19 pp × 4 years	≈	2020 Census boundary gap
0.75 years		**0.8 years**

The implied magnitude is close to the observed schooling discontinuity

Magnitude check, not an accounting exercise: the calculation assumes the marginal 19% would obtain four additional years of university.

Back: Results ◂ Back

Making the historical divide salient changes how east-side residents explain dropout

- Open question: why might a young person from your *colonia* not continue after high school?

Back

Making the historical divide salient changes how east-side residents explain dropout

- Open question: why might a young person from your *colonia* not continue after high school?

- Randomized boundary prime
  - first asked how often they hear *de la Calzada para allá*
  - nearly 80% hear or use the expression

Back

Making the historical divide salient changes how east-side residents explain dropout

- Open question: why might a young person from your *colonia* not continue after high school?

- Randomized boundary prime
  - first asked how often they hear *de la Calzada para allá*
  - nearly 80% hear or use the expression

- Answers coded using a fixed codebook

Back

Making the historical divide salient changes how east-side residents explain dropout

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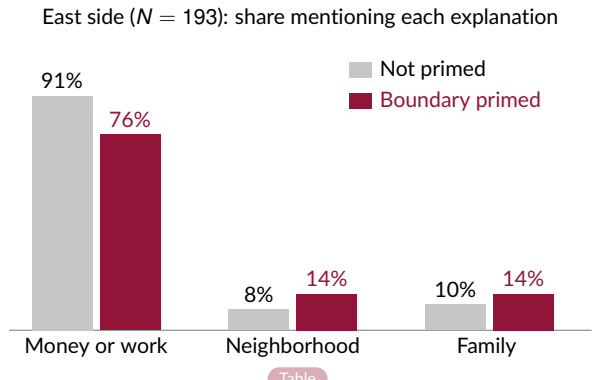
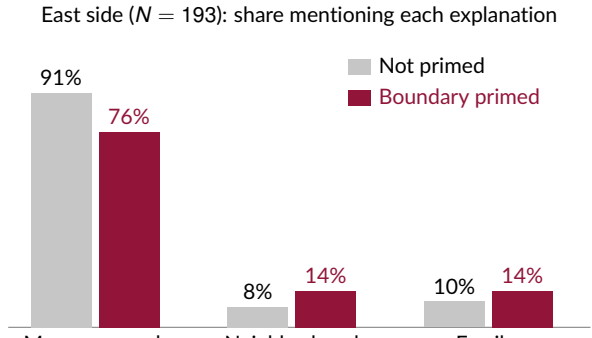


Making the historical divide salient changes how east-side residents explain dropout

- Open question: why might a young person from your *colonia* not continue after high school?
- Randomized boundary prime
  - first asked how often they hear *de la Calzada para allá*
  - nearly 80% hear or use the expression
- Answers coded using a fixed codebook



Table

Explanations based on money or work fall by **15 pp**; mentions of neighborhood and family both rise

Back

East-side households exchange more with neighbors, but value education the same

Reciprocal exchange and neighborhood orientation

	Mean west side	Mean east	East coefficient	q
Panel A. Localized prosociality (allocation exercises)				
Donation to a good cause	344.6	371.8	23.0 (24.3)	0.16
Allocation to a barrio stranger	244.6	262.9	19.7** (9.0)	0.06
Gives strictly more to the barrio stranger	0.000	0.166	0.170*** (0.032)	0.00
Allocation to a city stranger	251.1	265.6	14.6 (12.0)	0.13
Relative barrio premium	-0.015	0.053	0.088** (0.044)	0.07
Panel B. Localized network benefits and the cost of leaving				
Times helped by neighbors, past 12 months	0.97	1.96	1.10** (0.44)	0.07
Share of non-market time with neighbors	0.041	0.073	0.040** (0.018)	0.07
Share of non-market time supporting children's studies	0.333	0.319	−0.015 (0.044)	0.78
Vignette: the stayer's family gets more help	0.52	0.60	0.17 (0.23)	0.63
Vignette: leaving for university strains relations	0.59	0.72	0.16 (0.25)	0.63

East: help from neighbors about twice as often; parents who get more help demand less tutoring

Survey sample, $N = 238$

Income, schooling, roots | Help and tutoring | ◂ Back

East-side households exchange more with neighbors, but value education the same

Educational beliefs and norms

	Mean west side	Mean east	East coefficient	q
Panel C. Aspirations and expectations				
Aspires to high school or more	1.000	0.990	−0.007 (0.007)	1.00
Aspires to university	0.935	0.948	0.027 (0.036)	1.00
Expects high school or more	0.957	0.953	0.010 (0.027)	1.00
Expects university	0.870	0.850	0.020 (0.049)	1.00
Panel D. Returns to schooling and investment threshold				
Expected earnings, high-school graduate†	12,924	12,690	−0.045 (0.081)	1.00
Expected earnings, university graduate†	24,174	24,503	0.021 (0.078)	1.00
Perceived return to university (uni. − HS)	11,250	11,812	1,051 (1,657)	1.00
Perceives excluded-side graduate earns less	0.20	0.23	0.091 (0.177)	1.00
Reservation salary to justify university†	31,043	36,363	0.166** (0.071)	0.10

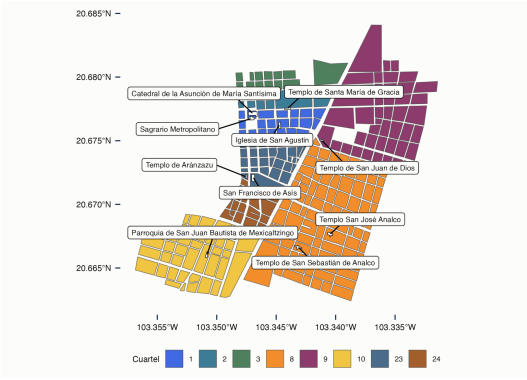
Same aspirations and perceived returns, but an **18% higher required salary** for university

Not a culture of poverty: preferences for education do not differ

Survey sample, $N = 238$

Returns to schooling | Norms and attachment: no difference | Not a culture of poverty | ◂ Back

Historical evidence of locally organized social life



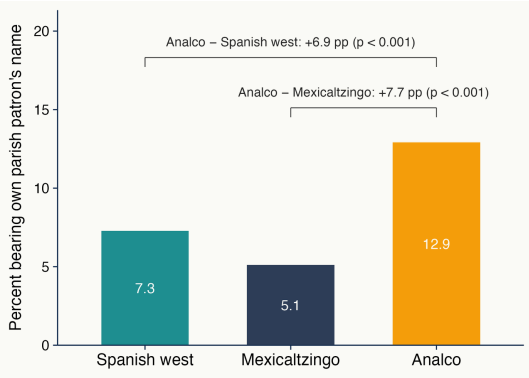
Parishes and patron saints around the creek

- **1821 census**
  given name = patron saint of one's own parish (Anderson and Spike, 2007)

Descriptive evidence, not a measure of local interdependence

Back

Historical evidence of locally organized social life

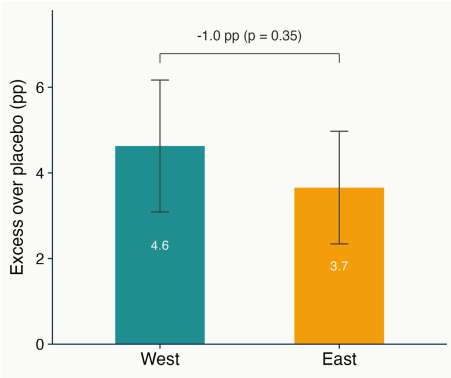


- **1821 census**
  given name = patron saint of one's own parish (Anderson and Spike, 2007)
- **Analco stands out relative to both western settlements**
  Mexicaltzingo was also Indigenous, but differed in colonial standing

Descriptive evidence, not a measure of local interdependence

Back

Historical evidence of locally organized social life



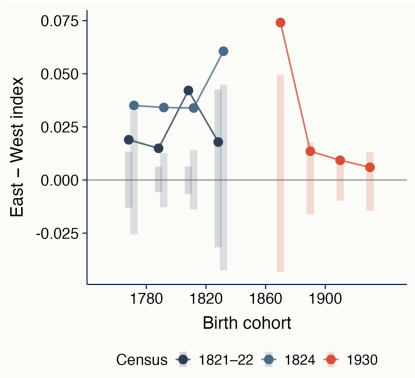
Named after the saint of the baptism day: ∼4 pp above a shifted calendar, on both banks

Descriptive evidence, not a measure of local interdependence

- **1821 census**
  given name = patron saint of one's own parish (Anderson and Spike, 2007)
- **Analco stands out relative to both western settlements**
  Mexicaltzingo was also Indigenous, but differed in colonial standing
- **Not religious naming generally**
  calendar-saint naming is similar across banks

Back

Historical evidence of locally organized social life



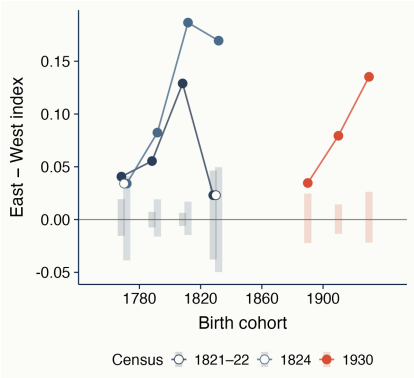
Given names: leave-one-out probability of living east given one's name, by birth cohort, within census

Descriptive evidence, not a measure of local interdependence

- **1821 census**
  given name = patron saint of one's own parish (Anderson and Spike, 2007)
- **Analco stands out relative to both western settlements**
  Mexicaltzingo was also Indigenous, but differed in colonial standing
- **Not religious naming generally**
  calendar-saint naming is similar across banks
- **Given names converged**
  no gap for cohorts born after 1880

Back

Historical evidence of locally organized social life



Paternal surnames: the same index; hollow points are not significant at the 5% level

Descriptive evidence, not a measure of local interdependence

- **1821 census**
  given name = patron saint of one's own parish (Anderson and Spike, 2007)
- **Analco stands out relative to both western settlements**
  Mexicaltzingo was also Indigenous, but differed in colonial standing
- **Not religious naming generally**
  calendar-saint naming is similar across banks
- **Given names converged**
  no gap for cohorts born after 1880
- **Surnames stayed distinct**
  they still differ in the 1930 census

Back

The 1992 explosions: positive selection, then a return to the neighborhood level

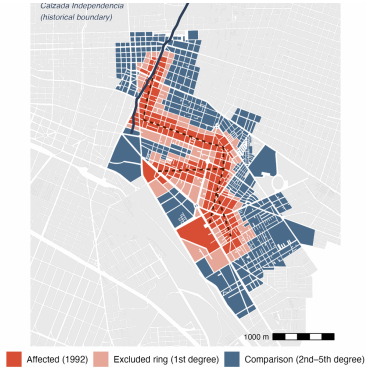
- Explosions destroyed part of Analco in 1992

Analco, 1992

Dynamic model ◂ Back

The 1992 explosions: positive selection, then a return to the neighborhood level

- Explosions destroyed part of Analco in 1992
- Rebuilding brought **new residents**

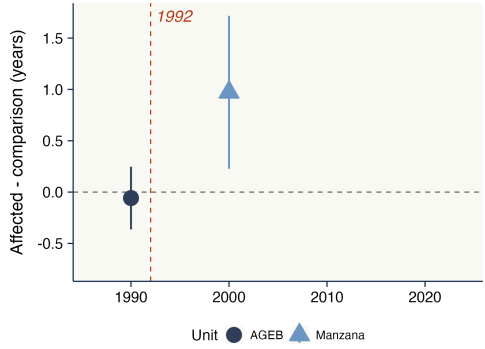


Rebuilt blocks within Analco

Dynamic model
Back

The 1992 explosions: positive selection, then a return to the neighborhood level

- Explosions destroyed part of Analco in 1992
- Rebuilding brought **new residents**
- Rebuilt blocks: **more schooled adults**

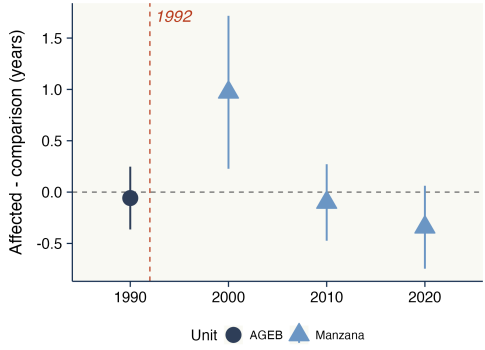


Adult schooling, rebuilt vs. neighboring blocks

Dynamic model ◂ Back

The 1992 explosions: positive selection, then a return to the neighborhood level

- Explosions destroyed part of Analco in 1992
- Rebuilding brought **new residents**
- Rebuilt blocks: **more schooled adults** at first

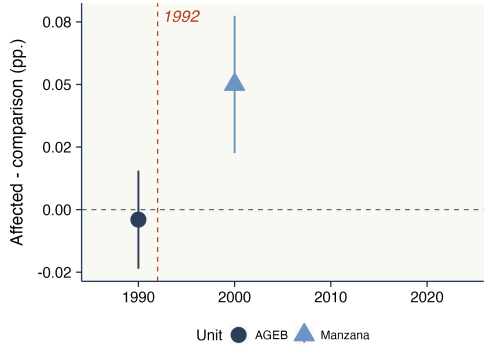


Adult schooling, rebuilt vs. neighboring blocks

Dynamic model ◂ Back

The 1992 explosions: positive selection, then a return to the neighborhood level

- Explosions destroyed part of Analco in 1992
- Rebuilding brought **new residents**
- Rebuilt blocks: **more schooled adults** at first
- Child attendance rose

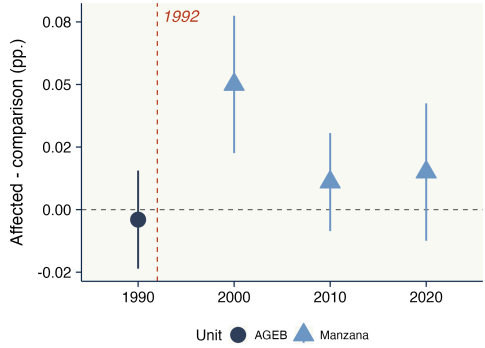


Child attendance, conditional on parental schooling

Dynamic model
Back

The 1992 explosions: positive selection, then a return to the neighborhood level

- Explosions destroyed part of Analco in 1992
- Rebuilding brought **new residents**
- Rebuilt blocks: **more schooled adults** at first
- Child attendance rose, then returned to the neighborhood level



Child attendance, conditional on parental schooling

The gap responds to who lives there: consistent with turnover and local adjustment

Descriptive: positive initial selection; not a test of the model

Dynamic model ◂ Back

Taking stock

- **Bringing the neighborhood to mind lowers demand for tutoring, on the east side only**
- **East-side households exchange more with neighbors, with the same aspirations and returns**
- **Social life east of the creek was locally organized already in 1821**

Exchange with neighbors, not a lower valuation of education

Local interdependence is not a culture of poverty

	Culture of poverty (Lewis, 1966)	**Local interdependence**	**Data**
Why less investment?	lower valuation of education	education puts the exchange with neighbors at risk	same aspirations and perceived returns
Responds to salience?	no: a stable trait	yes, where the exchange matters	yes, on the east side only
Who invests less?	the poorest, least organized	those most engaged with neighbors	more help, less tutoring; not the poorest
What is transmitted?	values, regardless of opportunities	participation, whose value depends on outside options	gap narrowed over the century
Social organization?	little beyond the family	dense among neighbors	more help, more trust in neighbors

◂ Back to ethnography ◂ Back to interpretation

Beyond Guadalajara, a neighborhood trust premium goes with less schooling

World Values Survey, wave 7: neighborhood trust premium = trust in neighbors relative to close ties

	Years of Education			Dropout Before Upper Secondary		
	(1)	(2)	(3)	(5)	(6)	(7)
Neighborhood trust premium (vs. close ties)	-0.018***	-0.025***	-0.022***	0.014***	0.019***	0.014***
	(0.004)	(0.004)	(0.004)	(0.004)	(0.004)	(0.004)
Out-group trust		0.051***	0.047***		-0.041***	-0.034***
		(0.004)	(0.005)		(0.005)	(0.005)
Trust in close ties (family + known)			0.011**			-0.018***
			(0.005)			(0.005)
Observations	47,072	46,952	46,952	47,072	46,952	46,952

35 countries; standardized outcomes; survey weights; robust SEs; region×year, region×settlement-size, gender×age, birth-country, ethnic, religious, and parental-education FE

Descriptive: stronger where ethnic groups are more segregated. Geographically local ties (Facebook SCI) in backup

Regions U.S. ◂ Back

Beyond Guadalajara, a neighborhood trust premium goes with less schooling

World Values Survey, wave 7: neighborhood trust premium = trust in neighbors relative to close ties

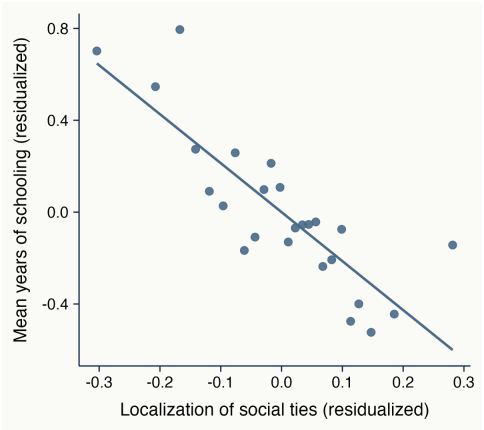
	Years of Education				Dropout Before Upper Secondary			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Neighborhood trust premium (vs. close ties)	-0.018***	-0.025***	-0.022***		0.014***	0.019***	0.014***	
	(0.004)	(0.004)	(0.004)		(0.004)	(0.004)	(0.004)	
Out-group trust		0.051***	0.047***			-0.041***	-0.034***	
		(0.004)	(0.005)			(0.005)	(0.005)	
Trust in close ties (family + known)			0.011**				-0.018***	
			(0.005)				(0.005)	
Neighborhood trust premium × Low ethnic segregation				-0.011**				0.005
				(0.005)				(0.005)
Neighborhood trust premium × High ethnic segregation				-0.038***				0.027***
				(0.007)				(0.008)
Observations	47,072	46,952	46,952	46,952	47,072	46,952	46,952	46,952

35 countries; standardized outcomes; survey weights; robust SEs; region×year, region×settlement-size, gender×age, birth-country, ethnic, religious, and parental-education FE

Descriptive: stronger where ethnic groups are more segregated. Geographically local ties (Facebook SCI) in backup

Regions U.S. ◂ Back

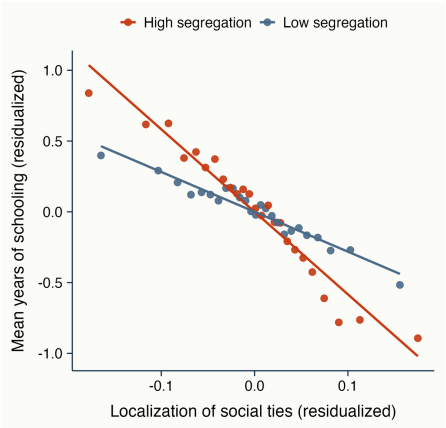
Across regions of the same country, more localized ties go with less schooling



Binned scatter of subnational regions. Localization of the Facebook friendship network (Bailey et al., 2018); schooling from the Global Data Lab. Both axes residualized on country FE, log population and log area.

Back: Beyond Guadalajara ◂ Back

Across U.S. ZIP codes, the gradient is about twice as steep in segregated counties



Binned scatter of ZIP codes within county; schooling from the ACS; same controls. Counties in the bottom and top terciles of the white/non-white residential dissimilarity index.

Back: Beyond Guadalajara ◂ Back

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